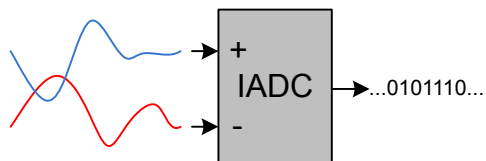
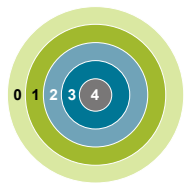


23. IADC - Incremental Analog to Digital Converter



Quick Facts

What?

The IADC is used to convert analog voltages into a digital representation and features high-speed, low-power operation.

Why?

In many applications there is a need to measure analog signals and record them in a digital representation, without exhausting the energy source.

How?

The low power IADC samples one or more input channels in a programmable sequence. With the help of PRS and DMA, the IADC can operate without CPU intervention in EM2 and EM3, minimizing the number of powered up resources. The IADC can be automatically shut down between conversions to further reduce the energy consumption.

23.1 Introduction

The IADC uses an Incremental Successive Approximation Architecture, with a resolution of up to 12 bits when operating at two million samples per second (2 Msps). The flexible incremental architecture uses oversampling to allow applications to trade speed for higher resolution. A high-accuracy mode enables greater than 15 bits of noise-free resolution. An integrated input multiplexer can select from external I/Os and several internal signals.

23.2 Features

- Flexible oversampled architecture allows for tradeoffs between speed and resolution.
 - Normal Mode
 - 1 Msps with oversampling ratio = 2
 - 76.9 ksps with oversampling ratio = 32
 - High Speed Mode
 - 2 Msps with oversampling ratio = 2
 - 153.8 ksps with oversampling ratio = 32
 - High Accuracy Mode
 - 10.7 ksps with oversampling ratio = 92
 - 3.8 ksps with oversampling ratio = 256
- Digital post-averaging
- Internal and external conversion trigger sources
 - Immediate (software triggered)
 - Local IADC timer
 - External TIMER module (synchronous with output / PWM generation)
 - General PRS hardware signal
- Integrated prescaler for conversion clock generation
- Can be run during EM2 and EM3, waking up the system on interrupts as needed
- Selectable reference sources
 - 1.21 V internal reference
 - External precision reference
 - Analog supply
- Support for offset and gain calibration
- Programmable input gain: 0.5x, 1x, 2x, 3x, or 4x
- Flexible output formatting
 - Unipolar or 2's complement bipolar data
 - Results can be saved in 12 bit, 16 bit, or 20 bit format
 - Programmable left or right justification
 - Optional channel ID tag
- Digital window comparison function detects when results are inside/outside a programmable window
- Two independent groups of configuration registers for setting IADC mode, clock prescaler, reference selection, oversample rate, unipolar/bipolar output formatting, and analog gain
- Programmable single channel conversion
 - Can use either configuration group
 - Triggered by any conversion trigger source
 - Can be tailgated after a scan sequence
 - One shot or continuous mode
 - Local FIFO for immediate data storage
 - Programmable watermark level to generate interrupt or initiate DMA transfer
 - Supports overflow and underflow interrupt generation
 - Supports window compare function
- Autonomous multi-channel scan
 - Up to 16 configurable slots in scan sequence
 - Each slot allows independent selection of configuration group, channel selection, and window compare enable
 - Triggered by any conversion trigger source
 - One shot or continuous mode
 - Local FIFO for immediate data storage
 - Programmable watermark level to generate interrupt or initiate DMA transfer
 - Supports overflow and underflow interrupt generation
 - Conversion tailgating support for predictable periodic scans

- Available interrupt sources:
 - Single FIFO has DVL (data valid level) entries available (also generates DMA request)
 - Scan FIFO has DVL (data valid level) entries available (also generates DMA request)
 - Single FIFO result compared true for digital compare window
 - Scan FIFO result compared true for digital compare window
 - Single queue conversion has completed
 - Scan queue entry conversion has completed
 - Scan queue table conversion has completed
 - Single FIFO overflow or underflow
 - Scan FIFO overflow or underflow
 - Polarity Error interrupt
 - Port Allocation Error interrupt
 - EM23 clock configuration error

23.3 Functional Description

The incremental ADC module block diagram is shown in [Figure 23.1 IADC Overview on page 747](#).

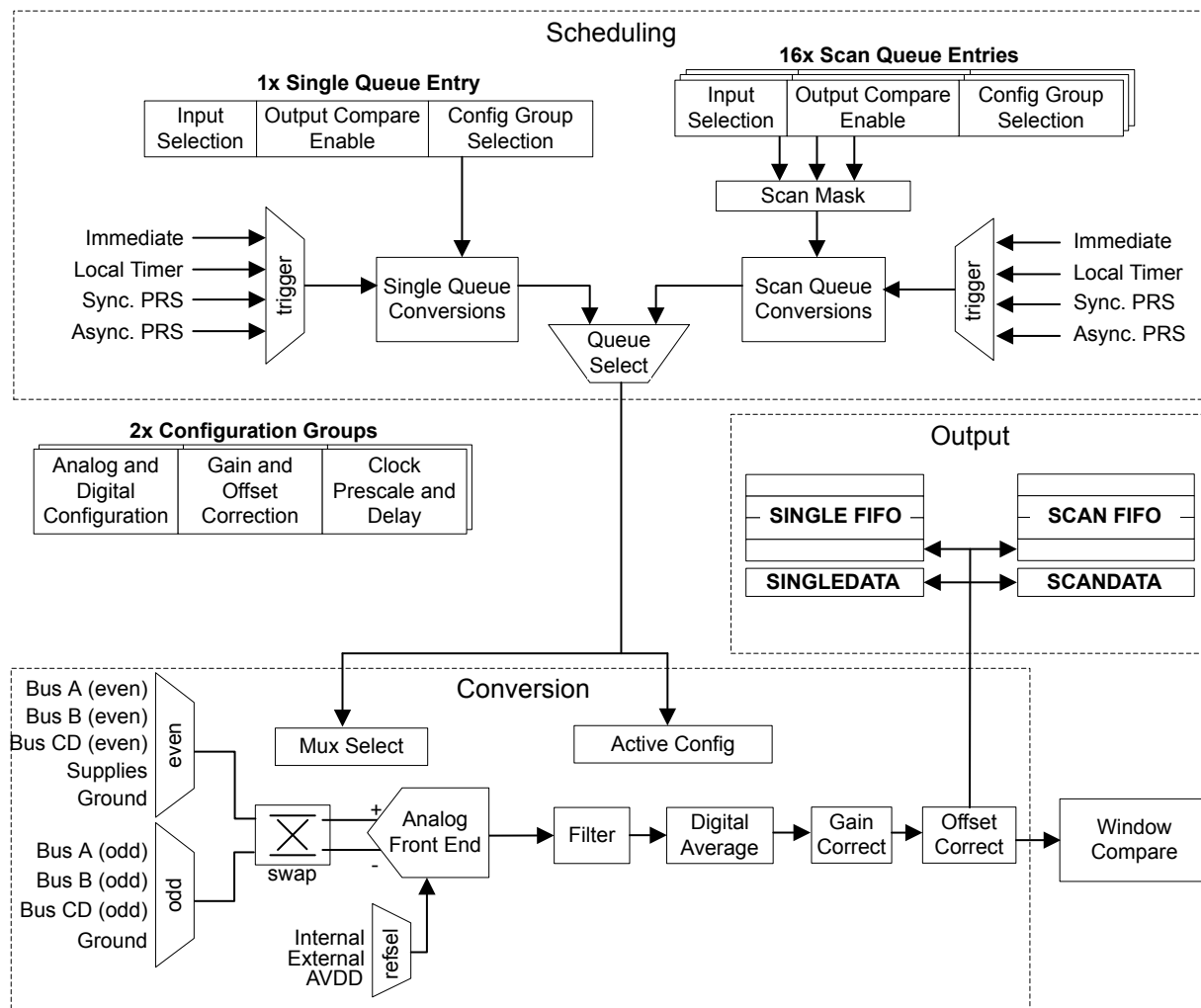


Figure 23.1. IADC Overview

23.3.1 Register Access

Many of the IADC module's configuration registers can only be written while the module is disabled (`IADC_EN_EN = 0`). These are `IADC_CTRL`, `IADC_TIMER`, `IADC_CMPTHR`, `IADC_TRIGGER`, `IADC_CFGx`, `IADC_SCALEx`, `IADC_SCHEx`, and `IADC_SCANx`. A typical setup sequence for the IADC module is:

1. With the IADC disabled (`IADC_EN_EN = 0`), program all configuration registers listed above.
2. Enable the IADC by setting `EN` in `IADC_EN` to 1.
3. Program the remaining configuration registers.
4. Enable the single or scan queue.
5. The IADC is ready for use.

23.3.2 Clocking

The IADC logic is partitioned into two clock domains: CLK_BUS (APBIF) and CLK_SRC_ADC (CORE). The APBIF domain contains the IADC registers and FIFO read logic. The rest of the IADC is clocked mainly by CLK_SRC_ADC and ADC_CLK, both of which are derived from CLK_CMU_ADC, as shown in .

CLK_CMU_ADC is the incoming clock routed to the ADC by the CMU, and may be up to 80 MHz. It is selected within the CMU module. If the ADC is to be used synchronously with an external TIMER module, the clock should be configured to derive from the group A clock. If configuring for operation in EM2 or EM3, a clock source available in EM2 and EM3 must be used directly, as the group A clock multiplexer will be shut down in EM2 and EM3.

CLK_SRC_ADC is derived from CLK_CMU_ADC, and must be no faster than 40 MHz. The HSCLKRATE field in IADC_CTRL sets the prescaler to divide CLK_CMU_ADC. If CLK_CMU_ADC is already 40 MHz or slower, HSCLKRATE can be set to 0x0 to pass the clock through to CLK_SRC_ADC without dividing it. CLK_SRC_ADC is the clock source used for the TIMEBASE prescaler as well as the local IADC timer.

ADC_CLK is used to drive the ADC front-end and state machine logic. Another prescaler is used to reduce CLK_SRC_ADC to a suitable frequency for the ADC operating mode. Different operational modes have different restrictions on the IADC clock speed. Because the operational mode may be different for single vs. scan conversions, or even for different conversions within a scan, each configuration group has a PRESCALE bit field in the IADC_SCHEx register.

When IADC_CFGx.ADCMODE is set to NORMAL, PRESCALE must be set to limit ADC_CLK to no faster than 10 MHz for 0.5x and 1x analog gain settings. For analog gain of 2x, 3x, and 4x, the maximum ADC_CLK is 5 MHz, 2.5 MHz, or 2.5 MHz respectively.

When IADC_CFGx.ADCMODE is set to HIGH SPEED, everything from NORMAL mode is scaled by a factor of 2. PRESCALE must be set to limit ADC_CLK to no faster than 20 MHz for 0.5x and 1x analog gain settings. For analog gain of 2x, 3x, and 4x, the maximum ADC_CLK is 10 MHz, 5 MHz, or 5 MHz respectively. It is recommended to run ADC_CLK no slower than 100 kHz.

When IADC_CFGx.ADCMODE is set to HIGH ACCURACY, PRESCALE must be set to limit ADC_CLK to no faster than 5 MHz, regardless of the analog gain setting.

These restrictions are summarized in [Table 23.1 Maximum ADC_CLK Speed vs Analog Gain and ADCMODE Settings on page 749](#).

Table 23.1. Maximum ADC_CLK Speed vs Analog Gain and ADCMODE Settings

Analog Gain	CFGx.ADCMODE = NORMAL	CFGx.ADCMODE = HIGH-SPEED	CFGx.ADCMODE = HIGH-ACCURACY
0.5 x	10 MHz	20 MHz	5 MHz
1 x	10 MHz	20 MHz	5 MHz
2 x	5 MHz	10 MHz	5 MHz
3 x	2.5 MHz	5 MHz	5 MHz
4 x	2.5 MHz	5 MHz	5 MHz

Note: If HSCLKRATE is configured to divide CLK_CMU_ADC by more than 1 (HSCLKRATE != 0), then PRESCALE must not be set to divide by 1 (PRESCALE = 0). When this condition is detected, a PRESCALE value of 1 (divide by 2) will be automatically be used instead of the programmed PRESCALE value.

The suspend mode fields IADC_CTRL_ADCCLKSUSPEND0 (for scan conversions) or IADC_CTRL_ADCCLKSUSPEND1 (for single conversions) can be used to shut down the clock between conversions and save power. The ADC logic will wake up the clock before starting IADC warmup and performing a conversion. If the suspend mode is set, the clock will shut down again once the conversion is complete.

When IADC_TRIGGER_SCANTRIGSEL or IADC_TRIGGER_SINGLETRIGSEL is set to IMMEDIATE, IADC_CTRL_ADCCLKSUSPENDn will force the clock to only be running when one of the queues is enabled.

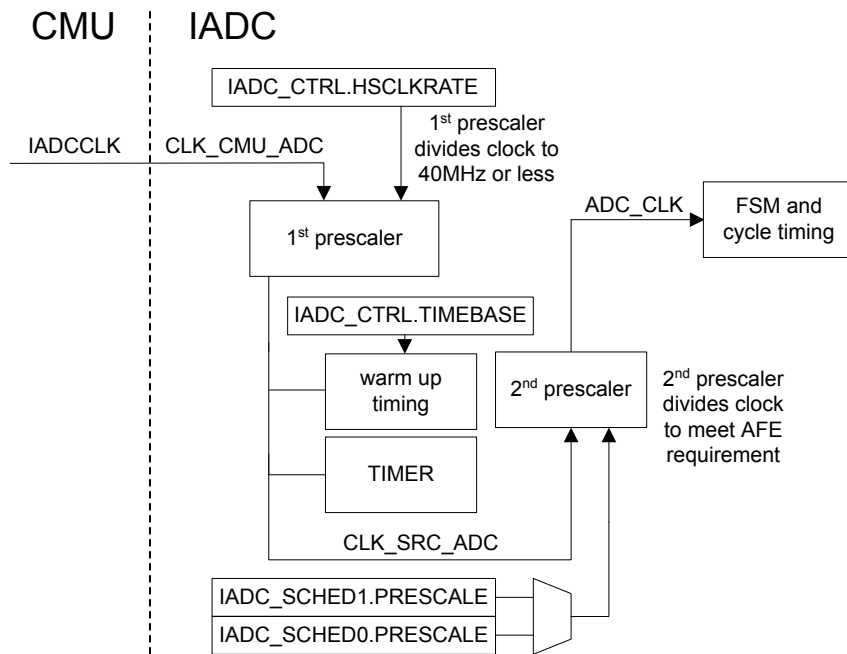


Figure 23.2. Clocking

23.3.3 Conversion Timing

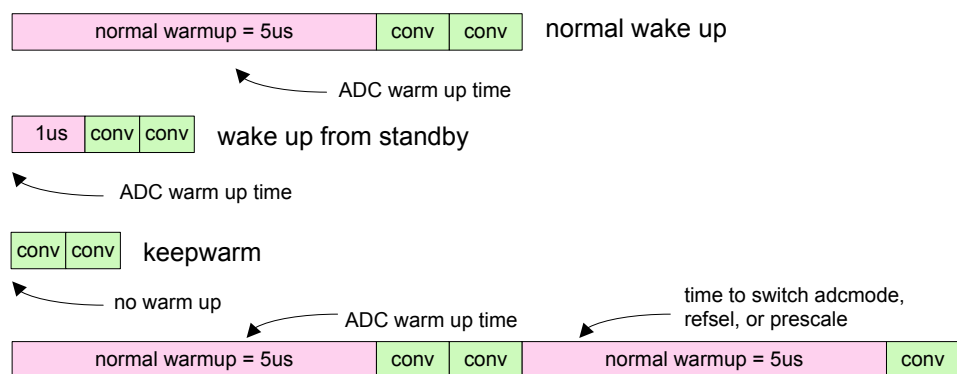
The IADC takes multiple samples of the analog signal to produce each output. The number of input samples contributing to an output word is determined by the oversampling ratio (OSR). Higher OSR settings will improve the ADC's INL and DNL, and reduce system-level noise, but require more time for each conversion. Different OSRs may be specified for each configuration group. It is important to note that oversampling is an analog process which provides more input samples to the digital filter. For Normal and High-Speed modes, the OSR is configured with the OSRHS bit field in the IADC_CFGx register. The OSR options for High-Accuracy mode are different, and are configured with the OSRHA bit field in the IADC_CFGx register.

During a conversion, the effective front-end sampling frequency (F_{sample}) in Normal and High-Speed modes is equal to $\text{ADC_CLK} / 4$. In High-Accuracy mode, F_{sample} is $\text{ADC_CLK} / 5$.

23.3.3.1 Warmup Time

To save energy, the IADC can be configured to power down completely or enter a standby state between conversions, if full speed operation is not required for the application. The required ADC warm up time from a full powered-down state is 5 μ s. Warmup from a standby state requires 1 μ s. Warmup is automatically timed by the ADC logic when it is required, but software must configure the TIMEBASE field in IADC_CTRL for a minimum 1 μ s interval. Note that the TIMEBASE counter receives CLK_SRC_ADC, and should be programmed based on that frequency. For example, if CLK_SRC_ADC is 40 MHz, TIMEBASE should be set to at least 0x27 (39) to produce the minimum 1 μ s interval. When transitioning from a powered-down state, the IADC will use five TIMEBASE intervals. When in standby the IADC will use one TIMEBASE interval.

The WARMUPMODE field in the IADC_CTRL register defines whether the IADC is powered down between conversions (WARMUPMODE = NORMAL), in standby between conversions (WARMUPMODE = KEEPINSTANDBY), or remains powered up (WARMUPMODE = KEEPWARM). The resulting start-up time is shown in [Figure 23.3 Start-up Timing on page 751](#). Note that even in WARMUPMODE = KEEPWARM or KEEPINSTANDBY, the ADC will implement 5 TIMEBASE intervals of warmup on initial power up, or any configuration change affecting PRESCALE, ADCMODE, or REFSEL. IADC_STATUS_ADCWARM reflects the current warmup status of the IADC.



Each change in ADCMODE, REFSEL, or PRESCALE require a 5us warm up period

Figure 23.3. Start-up Timing

23.3.3.2 Conversion Pipeline in Normal and High-Speed Modes

The IADC uses a pipelined architecture to perform different stages of the ADC conversion in parallel.

In Normal and High-Speed modes, the conversion time for a single sample can be determined from the OSR and the pre-scaled ADC_CLK frequency (f_{ADC_CLK}) as:

Conversion Time = $((4 * OSR) + 2) / f_{ADC_CLK}$

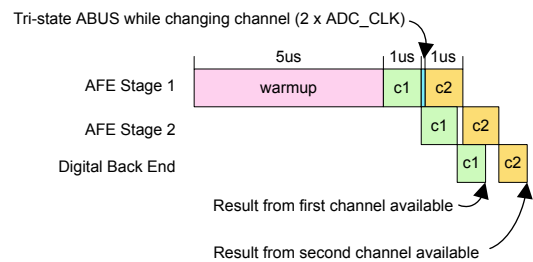
The minimum OSR is 2, meaning that the fastest possible conversion lasts 10 ADC_CLK clock cycles.

In Normal or High Speed mode, the IADC will automatically insert 2 additional cycles in the pipeline when changing channels to a new input. This allows for hold timing on the previous conversion and allows for time to tristate the ABUS analog buses before connecting the next input to the analog bus. Therefore the maximum sampling rate while continuously sampling on one channel in Normal mode (with ADC_CLK = 10 MHz) is 1 Msp/s, and the maximum sampling rate while switching channels is 833 ksp/s.

In High-Speed mode the allowed ADC_CLK speed is doubled to 20 MHz. The maximum sampling rate while continuously converting a single channel is 2 Msp/s, and the maximum sampling rate while switching channels is 1.67 Msp/s.

Figure 23.4 Normal Mode ADC Pipeline on page 752 and Figure 23.5 High-Speed Mode ADC Pipeline on page 753 show both single-channel and channel-switching scenarios powering up from a shutdown state with WARMUPMODE = NORMAL. The 5 us warm-up is shown in pink, a first conversion pipeline in green, and a second conversion in orange. The blue area in the top diagram represents the extra time to tristate while changing channels.

Normal mode switching channel between conversions



Normal mode converting same channel twice

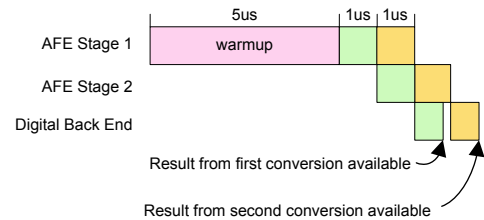
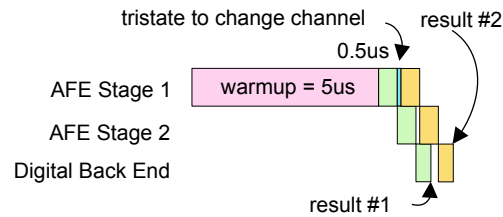


Figure 23.4. Normal Mode ADC Pipeline

High speed mode switching channel between conversions



High speed mode with same channel between conversions

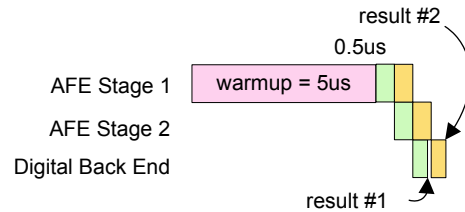


Figure 23.5. High-Speed Mode ADC Pipeline

23.3.3.3 Conversion Pipeline in High-Accuracy Mode

In High-Accuracy mode, the conversion time for a single sample can be determined from the OSR and the pre-scaled ADC_CLK frequency ($f_{\text{ADC_CLK}}$) as:

$$\text{Conversion Time} = ((5 * \text{OSR}) + 7) / f_{\text{ADC_CLK}}$$

The minimum OSR is 16, meaning that the fastest possible conversion in High-Accuracy mode lasts 87 ADC_CLK clock cycles.

In High-Accuracy mode, there are no additional cycles required to change channels to a new input. [Figure 23.6 High-Accuracy Mode ADC Pipeline on page 753](#) shows the pipeline timing when powering up from a shutdown state with WARMUPMODE = NORMAL. The 5 us warmup is shown in pink, a first conversion pipeline in green, and a second conversion in orange.

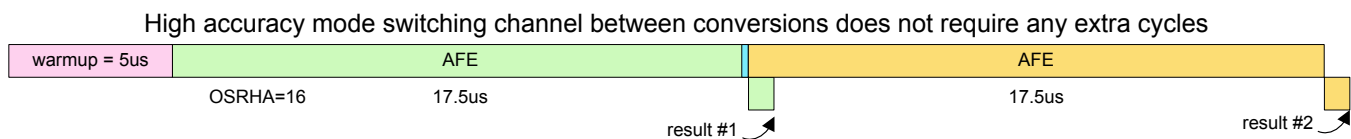


Figure 23.6. High-Accuracy Mode ADC Pipeline

23.3.3.4 Scheduling and Triggers

The IADC has several triggering options available for both the Single queue and the Scan queue. When a conversion trigger occurs and there are no other conversions active or pending, the request is serviced immediately. If both the single and scan queues are being used in an application, it is possible to serve the conversion requests as needed, and specify their priority.

Conversion triggering is configured using bit fields in the IADC_TRIGGER register. The SINGLETRIGSEL and SCANTRIGSEL fields specify the trigger source for Single and Scan conversion queues, respectively. The options for trigger source are:

- IMMEDIATE - Trigger from software. This is useful for triggering conversions on-demand from software with no specific sampling frequency requirements, or initiating continuous conversions at full speed.
- TIMER - Use the IADC local timer to trigger conversions. This is useful for triggering conversions at precise intervals.
- PRSCLKGRP - Use a synchronous PRS channel to trigger from an external peripheral in the same clock group domain (i.e. clock group A). This is useful for synchronizing conversions precisely with external TIMER events or PWM outputs.

Note: It is recommended to configure the PRS consumer registers prior to enabling synchronous PRS triggers to avoid false triggers.

- PRSPOS - Use a positive edge of an asynchronous PRS channel to trigger conversions. The trigger source will require 1-2 CLK_SRC_ADC cycles to synchronize. This is useful for triggering conversions as needed from asynchronous peripheral sources such as GPIO inputs, SYSRTC events, etc.
- PRSNEG - Use a negative edge of an asynchronous PRS channel to trigger conversions. This is the same as PRSPOS, but operates on negative edges of the selected input.

Both the single and scan trigger sources can be configured to generate one request per trigger, or begin continuous conversions. Setting SINGLETRIGACTION to ONCE will make one conversion request each time the selected single trigger occurs, and a single ADC output will be converted. Setting SINGLETRIGACTION to CONTINUOUS allows the single trigger to begin the first conversion, and when a conversion completes a new one will be requested immediately without requiring a new trigger. Channel selections and configuration should not be changed while SINGLETRIGACTION is set to CONTINUOUS. Doing so can produce conversion errors. The scan queue should be used if channel or configuration switching is required.

The SCANTRIGACTION field works to request conversion scans in a similar manner. Setting SCANTRIGACTION to ONCE will make one request each time the selected scan trigger occurs, and the IADC will perform all conversions specified in the scan once before stopping. Setting SCANTRIGACTION to CONTINUOUS allows the scan trigger to initiate continuous scans. When a scan cycle completes, a new one will be requested immediately without requiring a new trigger.

Conversion priority can be adjusted using the SINGLETAILGATE bit. By default, SINGLETAILGATE is set to TAILGATEOFF, meaning that conversion triggers are queued in the order they are received. Any conversion trigger for the Single queue or the Scan queue will initiate a conversion as soon as possible. If any conversion is already in progress or pending, the new conversion will be handled after the current operation.

Setting SINGLETAILGATE to TAILGATEON gives ultimate priority to the Scan queue. The IADC will only perform single conversions immediately after completion of a scan. This allows systems to use the scan queue for high-priority conversions with tight timing requirements, and the single queue for low-priority, on-demand conversion events. Note that this setting should only be used when scan conversions are guaranteed to trigger. If no scan sequence is triggered, any single conversion trigger will remain pending indefinitely. It is also important to note that if there is not enough time between scan conversions to service a single conversion, the next scan conversion will be delayed.

23.3.3.4.1 Conversion Triggering Examples

Scheduling a Single Sample

The simplest use case for the IADC is performing one conversion on-demand from the Single queue. [Figure 23.7 Immediate Single Conversion on page 755](#) shows the configuration and timing of this use case. The IADC warmup mode is configured for normal (shuts down between conversions). The single queue trigger is configured for immediate triggering of one conversion, and tailgating is turned off. When the conversion is requested (by setting IADC_CMD_SINGLESTART), the IADC block warms up and then begins converting. During the conversion, the CONVERTING bit in IADC_STATUS is set. When the conversion is complete, the queue is disabled, and SINGLEQEN returns low.

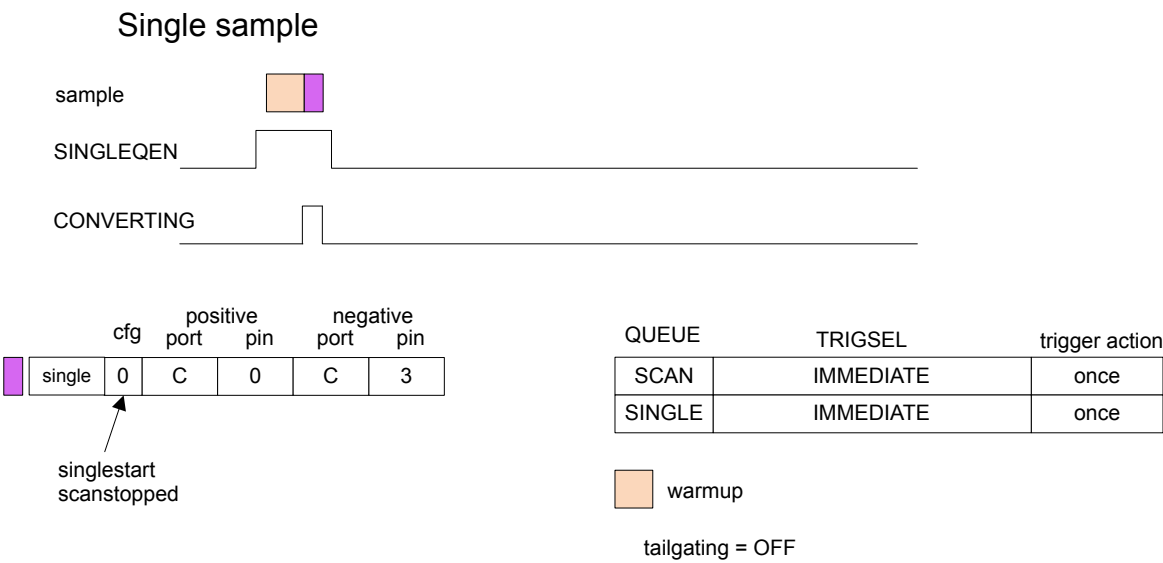


Figure 23.7. Immediate Single Conversion

Periodic Scans

Another common use case is to periodically trigger the IADC to perform a multi-channel scan. [Figure 23.8 Periodic Scan Example on page 756](#) shows the timing of a periodic scan triggered by the IADC's local timer. The scanner is configured to sample four different channels; two using configuration 0 and two using configuration 1. Note that a single TIMER trigger is used to initiate each scan, and all four samples are taken for each trigger. Note also that the IADC inserts another warmup time between conversions 1 and 2, when it switches from configuration 0 to configuration 1. The single queue is disabled and not used in this example.

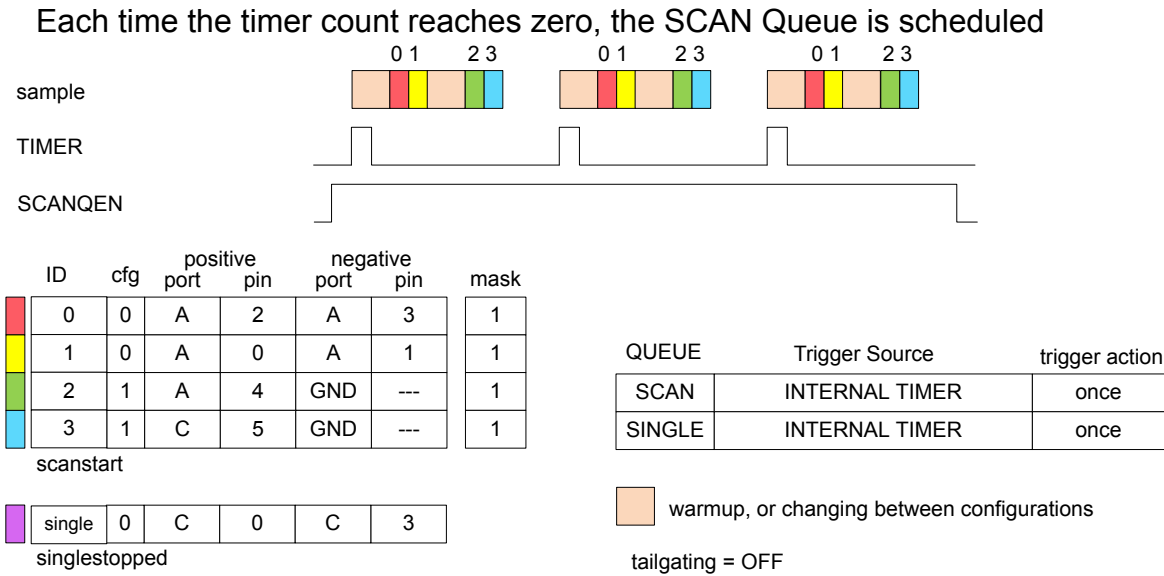


Figure 23.8. Periodic Scan Example

Tailgating Examples

An example using conversion tailgating is shown in [Figure 23.9 Simple Conversion with Tailgating Enabled on page 757](#). In the example, the Scan queue is configured to trigger a two-channel conversion periodically on the IADC local timer, while the Single queue is configured to trigger on-demand from software. When a single conversion is requested, it waits until after the scan sequence is complete, and then the single conversion is performed. The scan conversions are using configuration 0, and the single conversion is using configuration 1, so a warmup delay is inserted between the end of the scan and the beginning of the single conversion cycle. Note that this example provides plenty of time between IADC scan conversions for the single conversion to occur, and no scan conversions are delayed.

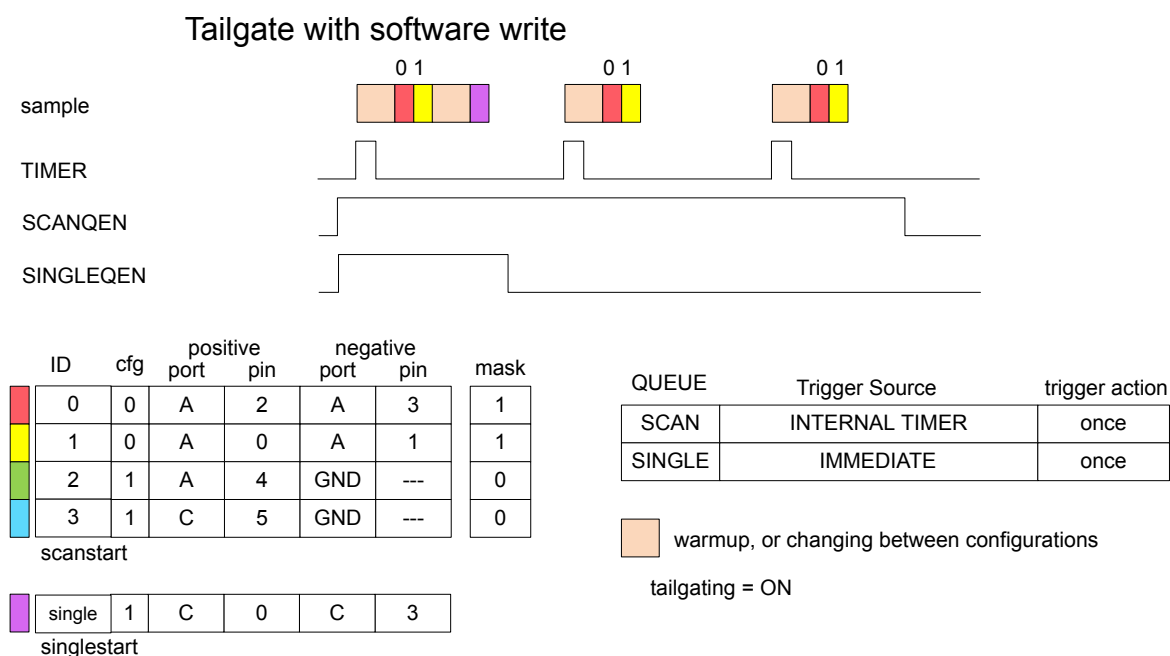


Figure 23.9. Simple Conversion with Tailgating Enabled

Another example, shown in [Figure 23.10 Conversions with Tailgating Disabled on page 758](#), demonstrates how requests are handled on the different conversion queues with tailgating disabled.

In this example, the scan queue is being triggered on the internal timer while the single queue is being triggered on a PRS positive edge. Since tailgating is not enabled, the queues will be serviced on a first come first served basis. The first single queue trigger falls between two scan queue triggers and does not interfere with scan queue timing. The second single queue trigger happens just before the scan queue trigger. The IADC will complete this single queue conversion and delay the next scan queue conversions.

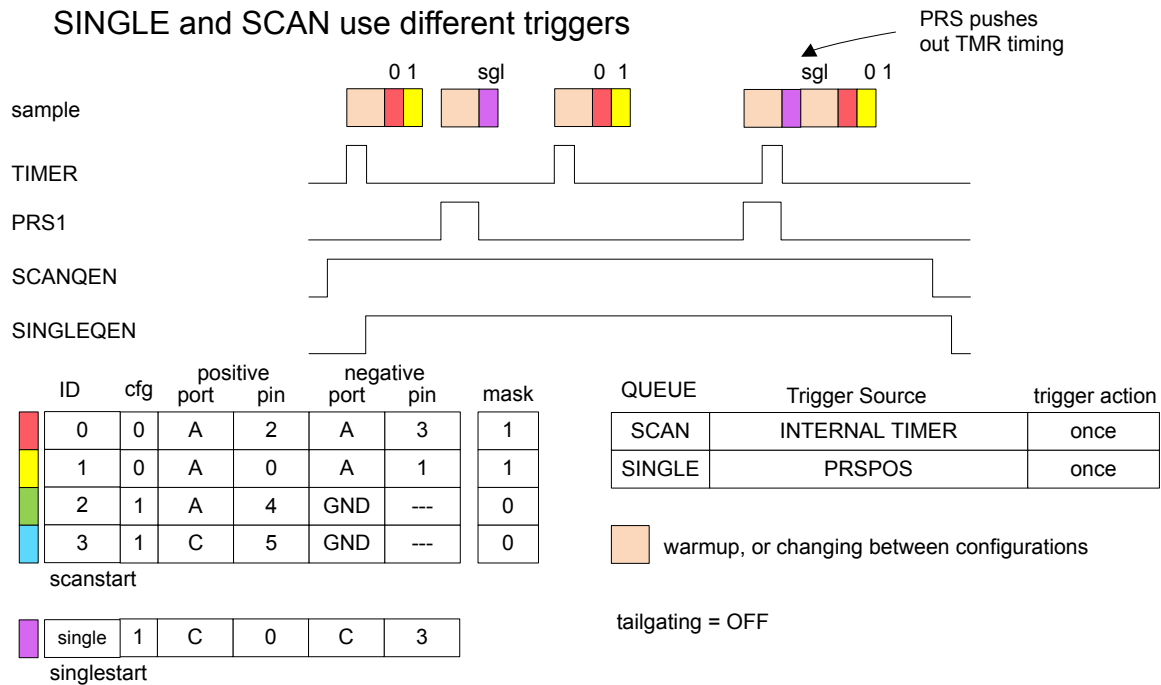


Figure 23.10. Conversions with Tailgating Disabled

Continuous Conversions

An example of continuous conversions triggered from the scan queue is shown in [Figure 23.11 Continuous Conversions on page 758](#). In this example the SCANTRIGACTION field in IADC_TRIGGER is set to CONTINUOUS, and the conversion trigger source is software (SCANTRIGSEL = IMMEDIATE). When the scan queue is enabled with IADC_CMD_SCANSTART, the ADC warms up and then performs repeated back-to-back scans until software disables the scan queue using IADC_CMD_SCANSTOP. While this example shows only one channel converted continuously, it is possible to enable multiple channels for the scan sequence.

Continuous

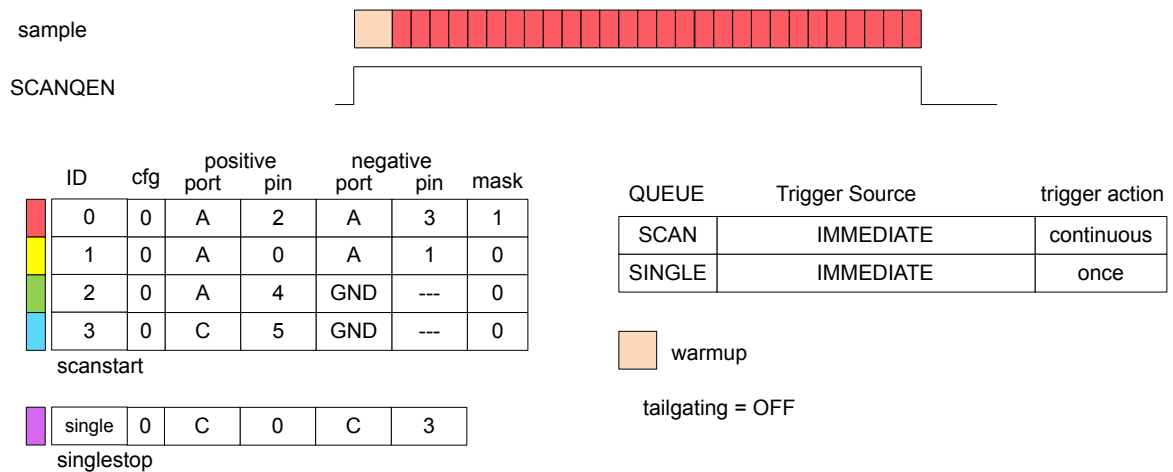


Figure 23.11. Continuous Conversions

23.3.4 Reference Selection and Analog Gain

The default IADC reference is to use the internal band gap circuit. The analog power supply voltage can also be used as a voltage reference. The reference voltage is selected using the REFSEL field in IADC_CFGx.

Table 23.2. Mode Settings

Reference	Description	Normal and High-Speed Mode Voltage Range	High-Accuracy Mode Voltage Range
VBGR	Internal	1.21V	1.21V
VDDX	Analog Power Supply	AVDD	AVDD
VREF	External	1.0V - AVDD (1.25V Nominal)	1.0V - 1.25V (1.25V Nominal)
VREF2P5	External	1.0V - AVDD (1.25V Nominal)	1.0V - 2.5V (2.5V Nominal)
Note: 1. In high-accuracy mode with VREF2P5 selected, the AVDD supply must be at least 3V.			

The IADC also has analog gain selection, controlled via the ANALOGGAIN field in IADC_CFGx. The analog gain can be set to 0.5x, 1x, 2x, 3x, or 4x. Note that 2x, 3x, and 4x gain modes may require slower ADC_CLK. The analog gain impacts where the full-scale input reading occurs. For example, with a 1.25 V external reference and ANALOGGAIN set to 2x, the analog input to the IADC is multiplied by a factor of 2, and a full-scale reading occurs at $1.25 \text{ V} / 2 = 0.625 \text{ V}$. If ANALOGGAIN is set to 0.5x, the full-scale reading of the ADC will not occur until the input reaches 2.5 V. Note that the ADC is only capable of measuring inputs within the supply rails of the device. If the full scale is configured to be greater than the supply voltage, the maximum input will be limited to the supply.

The sampling capacitance (C_{sample}) is changed according to the analog gain setting.

Table 23.3. Input Sampling Capacitance vs. Analog Gain

Analog Gain Setting	Input Sampling Capacitance
0.5x	1 pF
1x	2 pF
2x	4 pF
3x	6 pF
4x	8 pF

Given the sampling capacitance and the front-end sampling rate (F_{sample}), the input impedance of the converter can be calculated as:

$$Z_{\text{in}} = 1 / (C_{\text{sample}} * F_{\text{sample}})$$

Note that the input is not sampled when the converter is inactive between conversions and operating with WARMUPMODE = NORMAL or KEEPINSTANDBY with longer intervals between conversion triggers can increase the effective input impedance of the converter.

23.3.5 Input and Configuration Selection

The IADC supports measurement on a number of internal and external signals. External signals are routed to GPIO through shared ABUS resources on the device, or (on some devices) through dedicated analog inputs available to the IADC block.

The single queue and the scan queue have separate registers available to select inputs and configurations. The IADC_SINGLE register is used to select the input and configuration for the single queue. The IADC_SCANx registers are used to select the inputs and configurations for each of the scan table entries. In both cases, the register contents and setup are similar. The PORTPOS and PINPOS fields are used to select a signal for the positive ADC input, while PORTNEG and PINNEG are used to select a signal for the negative ADC input. The CFG field selects which of the two configuration sets will be used with the input (i.e. configuration options specified in IADC_CFGx, IADC_SCALEx, and IADC_SCHEx).

To perform single-ended conversions, the PORTNEG field should be set to GND. This indicates that the positive ADC input will be measured with reference to chip ground. PORTPOS and PINPOS should be used to select the desired input signal. The PINNEG field is not used for single-ended conversions.

To perform differential conversions, PORTPOS, PINPOS are used to select the positive input to the ADC, while PORTNEG and PINNEG are used to select the negative input. Note that there are two independent multiplexers in the ADC, and firmware cannot select two signals from the same multiplexer for a differential measurement. The "even" multiplexer consists of all EVEN ABUS selections, Supply voltage options, and GND. The "odd" multiplexer consists of all ODD ABUS selections and GND. One selection from each multiplexer is allowed on the positive and negative input. More detailed examples may be found in [23.3.5.3 ABUS Input Selection Examples](#).

The scan queue has one additional register, IADC_MASKREQ, to specify which of the 16 possible channel slots will be converted during a scan operation. Each channel in the scan queue is enabled by writing the corresponding bit in the IADC_MASKREQ register to 1. Enabled channels will be converted in sequence from lowest to highest, during a scan. See [23.3.5.4 Scan Queue](#) for more details on using the scan queue.

23.3.5.1 External GPIO Inputs

GPIO input selections are routed through shared ABUS resources. In order for the IADC to use any GPIO as an input, the IADC must be allocated appropriate analog bus resources in the GPIO_ABUSALLOC, GPIO_BBUSALLOC, or GPIO_CDBUSALLOC registers. For example, if IADC0 will be using both odd and even numbered pins on GPIO port PA, then AEVEN0 and AODD0 in GPIO_ABUSALLOC could both be set to IADC0. This gives IADC0 access to these two buses. Generally, bus access is set to specific peripherals at configuration time and left alone - it is not normally required to change the bus allocation on the fly. If the IADC requests a pin from a bus that has not been allocated to the IADC, an error will be generated, the PORTALLOCERRIF in IADC_IF will be set, and any conversion result will be 0. For more details on analog bus structure and capabilities, refer to the GPIO section.

When the appropriate analog buses have been configured to route to the IADC, GPIO selection is a simple matter of programming the desired port and pin into the PORTPOS, PINPOS, PORTNEG, and PINNEG fields. For example, to configure a channel to convert the differential voltage between pins PA5 and PA4, PORTPOS = PORTA, PINPOS = 5, PORTNEG = PORTA, PINNEG = 4. If an invalid selection is made, a polarity error will be generated. More specific examples are described in [23.3.5.3 ABUS Input Selection Examples](#).

23.3.5.2 Internal and Dedicated Inputs

Internal signals and dedicated inputs are not routed through the shared ABUS resources. In general, these resources are selected directly by the settings of PORTPOS and PORTNEG, while the PINPOS and PINNEG fields are not used. When PORTPOS is set to SUPPLY, PINPOS is used to select which of the power supplies is connected. To facilitate power supply measurements using internal reference options, higher voltage supplies are attenuated by a factor of 4.

Table 23.4. Supply Selection (PORTPOS = SUPPLY)

PINPOS	Supply Connection	Voltage at Positive Input
0	AVDD	AVDD / 4
1	IOVDD	IOVDD / 4
2	VSS	VSS
3	VSS	VSS
4	DVDD	DVDD / 4
7	DECOUPLE	DECOUPLE

If an internal signal is selected for PORTPOS or PORTNEG, selecting GND on the opposite input will instruct the converter to perform a single-ended conversion. In the case where PORTPOS = GND, the IADC logic will automatically swap the direct input selected by PORTNEG to the positive input of the ADC. Otherwise, a differential conversion is performed with PORTPOS selecting the positive and PORTNEG selecting the negative input.

23.3.5.3 ABUS Input Selection Examples

When configuring to measure a single-ended signal through the ABUS, the positive input selection should always point to the desired input, and PORTNEG should be programmed to GND.

Correct configuration examples for single-ended conversions are shown in [Figure 23.12 Single-Ended Port/Pin Selection Odd Channel on page 762](#) and [Figure 23.13 Single-Ended Port/Pin Selection Even Channel on page 762](#). Note that the IADC logic will automatically swap the appropriate multiplexer to the positive input of the ADC.

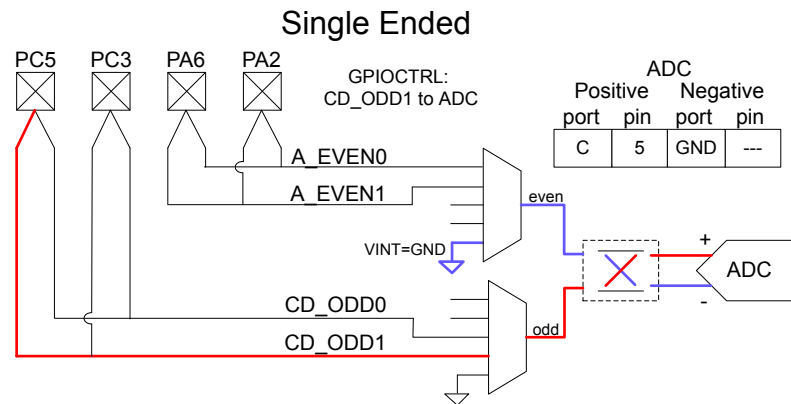


Figure 23.12. Single-Ended Port/Pin Selection Odd Channel

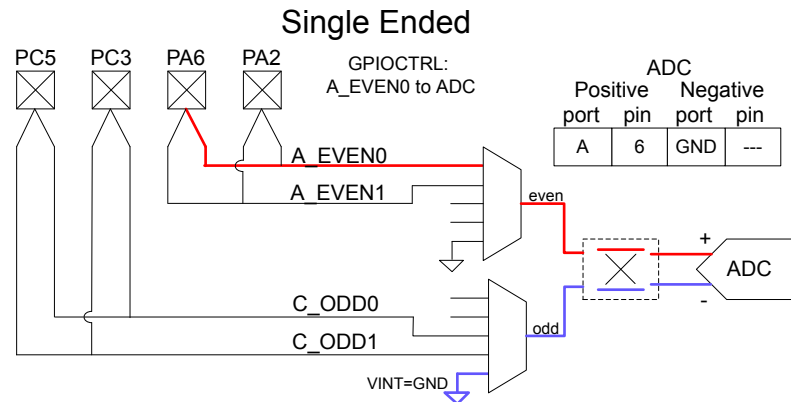


Figure 23.13. Single-Ended Port/Pin Selection Even Channel

[Figure 23.14 Single-Ended Port/Pin Selection Polarity Error on page 763](#) shows an example where the PORTPOS input has been configured to GND, with PORTNEG and PINNEG configured for a GPIO pin. This will result in a polarity error (POLARITYERRIF in IADC_IF will be set) and any conversion result will be 0.

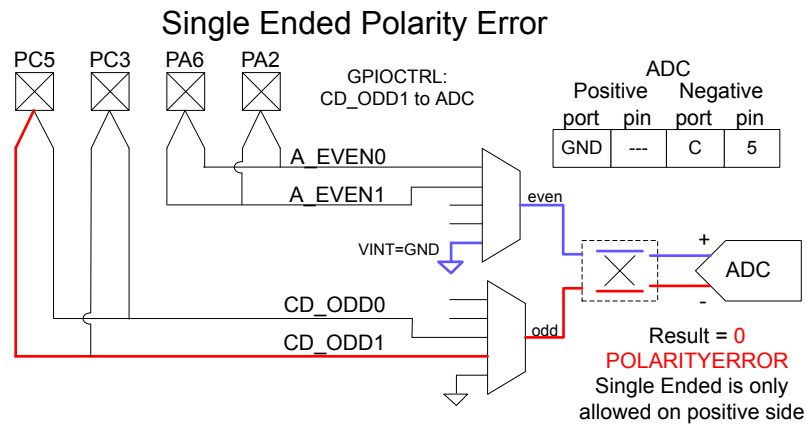


Figure 23.14. Single-Ended Port/Pin Selection Polarity Error

Correct configuration examples for differential conversions are shown in [Figure 23.15 Differential Port/Pin Selection without Swap on page 763](#) and [Figure 23.16 Differential Port/Pin Selection with Swap on page 764](#). In both these examples, the inputs were selected from one EVEN multiplexer channel and one ODD multiplexer channel. As with single-ended mode, the IADC logic will automatically swap the multiplexer connections to the IADC input if needed.

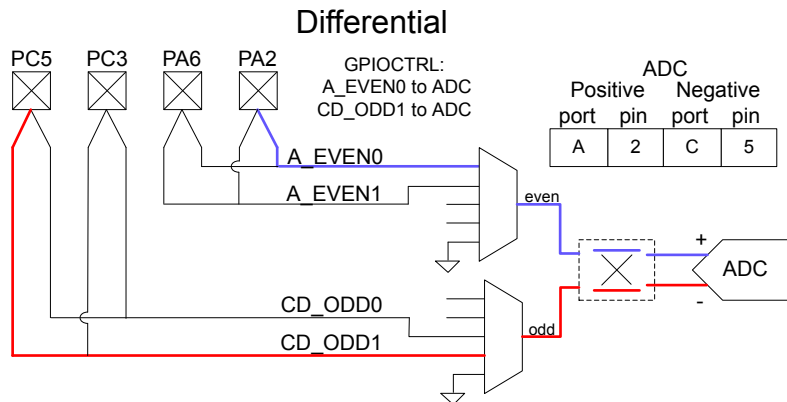


Figure 23.15. Differential Port/Pin Selection without Swap

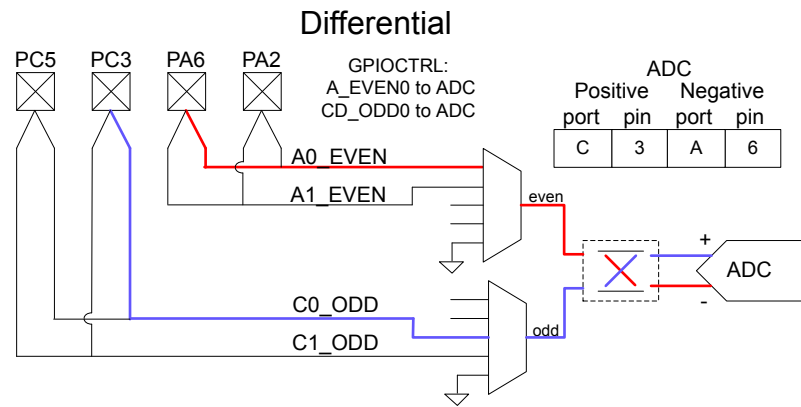


Figure 23.16. Differential Port/Pin Selection with Swap

Figure 23.17 Differential Port/Pin Selection Polarity Error on page 764 shows an example where the both the positive and the negative input selections point to ODD buses. Even though the IADC has been allocated both buses, they both route through the ODD input multiplexer and cannot be measured against one another. This will result in a polarity error (POLARITYERRIF in IADC_IF will be set) and any conversion result will be 0x7FFFF. Likewise, a polarity error will occur if both inputs are selected from EVEN buses.

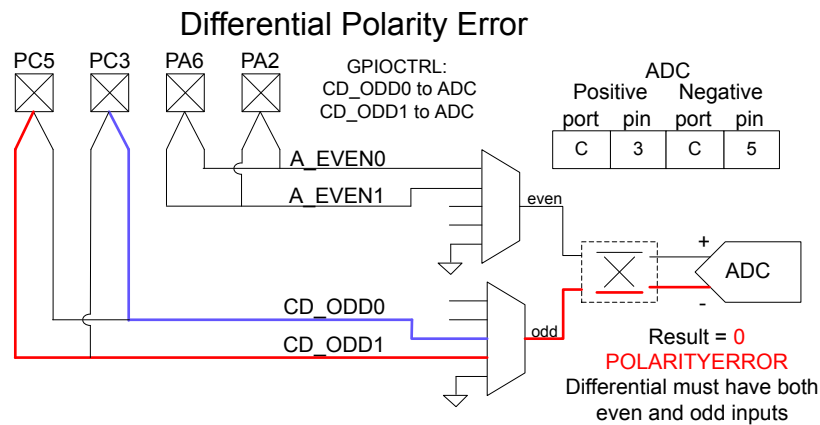


Figure 23.17. Differential Port/Pin Selection Polarity Error

23.3.5.4 Scan Queue

The scan queue allows the IADC to automatically convert up to 16 channels in sequence without CPU intervention. Input and configuration selection for each channel in the scan table is specified by the IADC_SCANx register for that channel (channel 0 is configured with IADC_SCAN0, channel 1 is configured with IADC_SCAN1, and so on). The IADC_MASKREQ register allows software to define which of the scan table entries (IADC_SCANx) to convert during a scan. For example, channels 0, 1, and 7 can be enabled by writing bits 0, 1, and 7 of IADC_MASKREQ to 1 (IADC_MASKREQ = 0x0083).

The IADC_SCANx registers must be configured when the IADC module is disabled (IADC_EN_EN = 0). IADC_MASKREQ can be written while IADC_EN_EN is set to 1. If a scan operation is in progress, MASKREQ will be synchronized and held until the current scan operation has completed. Then MASKREQ is copied into the STMASK register for the next scan operation. IADC_STMASK is the working copy of the MASKREQ used by the IADC during a scan. MASKREQ will only transfer to STMASK when the scan queue is not scanning and converting the scan table. IADC_STATUS_MASKWRITEPENDING can be used by software to see when the MASKREQ write has been transferred to STMASK. Writing a new MASKREQ in the middle of a scan will not corrupt the current scan. Software which writes to MASKREQ during a scan operation must ensure IADC_STATUS_MASKWRITEPENDING returns to 0 before updating IADC_MASKREQ again. [Figure 23.18 MASKREQ Updates on page 765](#) shows a time line of when the MASKREQ write is updated.

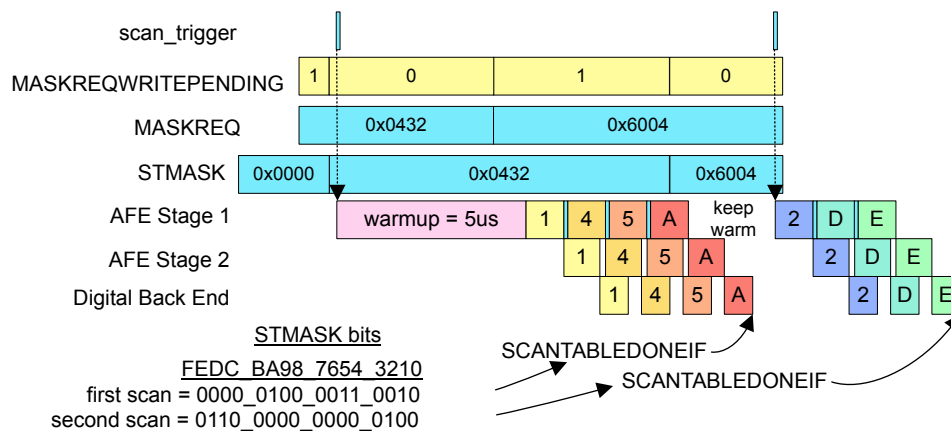


Figure 23.18. MASKREQ Updates

23.3.6 Gain and Offset Correction

The IADC has built in gain and offset correction capabilities. Each of the two configuration groups contains its own correction values stored in the IADC_SCALEx register, allowing the IADC to automatically apply the appropriate correction for the IADC configuration that is being used.

Gain correction is performed through a fixed-point 16-bit value with a range from 0.75x to 1.2499x. The 3 MSBs of the gain value are not directly writeable. The GAIN3MSB bit in IADC_SCALEx is used to select between 011 and 100 for the 3 MSBs, and the lower 13 bits are programmed directly into IADC_SCALEx_GAIN13LSB. Clearing GAIN3MSB to 0 selects the most significant bits of the gain as 011, representing a range from 0.75x to 0.9999x. Setting GAIN3MSB to 1 selects the most significant bits of the gain as 100, representing a range from 1.00x to 1.2499x.

Offset correction is controlled by the OFFSET field in IADC_SCALEx. It is important to note that the offset correction does not have a direct 1-to-1 relationship with the LSB of the IADC output, and depends on both the OSR and gain correction settings. The offset correction range is +/- 12.5% of full scale. OFFSET is encoded as a 2's complement, 18-bit number with the LSB representing $1 / 2^{20}$ of full scale. Thus, bit 8 of OFFSET aligns with bit 0 of the 12-bit IADC output word.

23.3.6.1 Using Production Calibration Parameters

IADC calibration is performed on every device during Silicon Labs production test and production calibration parameters are stored in the flash DI page. The production calibration values are useful for a wide variety of possible IADC configurations, but do not map directly to the offset and gain correction fields in the IADC_SCALEx registers. Software must calculate the actual offset and gain correction values from the factory calibration values.

23.3.6.1.1 Gain Correction

Gain error is measured during production test at various settings of ANALOGGAIN, and stored in the DEVINFO_IADC0GAIN0 and DEVINFO_IADC0GAIN1 locations. The GAINCANA1 field is used for 0.5x and 1x ANALOGGAIN settings, while GAINCANA2, GAINCANA3, and GAINCANA4 are used for ANALOGGAIN settings of 2x, 3x, and 4x, respectively.

The GAINCANAn values are expressed as the full 16-bit fixed-point gain, and must be compressed before writing to the IADC_SCALEx register.

23.3.6.1.1.1 Gain Correction in Normal / High Speed Mode

When the IADC is operated in Normal / High Speed mode, a 1st order filter is employed in the decimation. The nominal gain value in these modes for all OSRHS settings is 1.0, or 0x8000 as expressed in the fixed-point 16-bit format. The IADC gain error is designed to be minimal with the digital gain correction set to 1.0 (GAIN3MSB = 1 and GAIN13LSB = 0). Tighter gain error is achieved by adjusting these values in IADC_SCALEx. Using this gain correction mechanism will result in a slight increase to the DNL of the converter, which is reduced by higher OSR settings.

To apply a factory-calibrated gain:

1. Read the appropriate GAINCANAn field from the DEVINFO locations for the selected ANALOGGAIN.
2. Write the MSB (bit 15) of GAINCANAn to GAIN3MSB in IADC_SCALEx.
3. Write the 13 LSBs (bits 12-0) of GAINCANAn to GAIN13LSB in IADC_SCALEx.

23.3.6.1.1.2 Gain Correction in High Accuracy Mode

When the IADC is operated in High Accuracy mode, a 2nd order filter is employed in the decimation. The nominal gain value of the filter is dependent on the OSRHA setting. The gain value stored in DEVINFO space must be adjusted before applying to the IADC_SCALEx register.

To apply a factory-calibrated gain:

1. Read the appropriate GAINCANAn field from the DEVINFO locations for the selected ANALOGGAIN.
2. Multiply the value by the OSR gain correction factor (ha_gain) found in [Table 23.5 Ideal High Accuracy Gain Correction on page 766](#).
3. Write the MSB (bit 15) of the result to GAIN3MSB in IADC_SCALEx.
4. Write the 13 LSBs (bits 12-0) of the result to GAIN13LSB in IADC_SCALEx.

Table 23.5. Ideal High Accuracy Gain Correction

OSRHA Setting	OSR	16-bit OSR Gain Correction (ha_gain)
HIACC16	16 x	0x7879
HIACC32	32 x	0x7C1F
HIACC64	64 x	0x7E08
HIACC92	92 x	0x7A8E
HIACC128	128 x	0x7F02
HIACC256	256 x	0x7F80

23.3.6.1.2 Offset Correction

Offset is impacted by the selected ANALOGGAIN and OSR settings in IADC_CFGx, the GAIN3MSB and GAIN13LSB values in IADC_SCALEx, and the voltage reference. Offset is production calibrated for any combination of possibilities, but the OFFSET register value must be calculated for the given situation before it can be effectively used.

23.3.6.1.2.1 Offset Correction in Normal Mode

The production offset calibration consists of four 16-bit terms written to the DEVINFO space: OFFSETANA1NORM, OFFSETANA2NORM, OFFSETANA3NORM, and OFFSETANABASE. The following procedures will determine the setting for the OFFSET register based on production calibration values.

Step 1: Determine the offset gain adjustment term (off_gain) based on ANALOGGAIN.

For ANALOGGAIN set to 0.5x or 1x:

$$\text{off_gain} = 0$$

For ANALOGGAIN set to 2x, 3x, or 4x, off_gain is calculated as:

$$\text{off_gain} = \text{OFFSETANA2NORM} * (\text{gain} - 1)$$

The following table summarizes these equations:

Table 23.6. Offset Gain Adjustment

ANALOGGAIN Setting	Analog front-end gain	Offset Gain Adjustment Term (off_gain)
ANAGAIN0P5	0.5 x	0
ANAGAIN1	1 x	0
ANAGAIN2	2 x	OFFSETANA2NORM * 1
ANAGAIN3	3 x	OFFSETANA2NORM * 2
ANAGAIN4	4 x	OFFSETANA2NORM * 3

Step 2: Calculate the analog offset adjustment term (off_ana) based on OSR and off_gain.

For an OSR of 2x (OSRHS = 0):

$$\text{off_ana} = \text{OFFSETANA1NORM} + \text{off_gain}$$

For all other OSR settings, 4x - 64x:

$$\text{off_ana} = \text{OFFSETANABASE} + 2 * (\text{OFFSETANA3NORM} - \text{off_gain}) / \text{OSR}$$

The following table expresses these equations:

Table 23.7. Analog Offset Adjustment

OSRHS Setting	OSR	Analog Offset Adjustment Term (off_ana)
HISPD2	2 x	OFFSETANA1NORM + off_gain
HISPD4	4 x	OFFSETANABASE + (OFFSETANA3NORM - off_gain)/2
HISPD8	8 x	OFFSETANABASE + (OFFSETANA3NORM - off_gain)/4
HISPD16	16 x	OFFSETANABASE + (OFFSETANA3NORM - off_gain)/8
HISPD32	32 x	OFFSETANABASE + (OFFSETANA3NORM - off_gain)/16
HISPD64	64 x	OFFSETANABASE + (OFFSETANA3NORM - off_gain)/32

Step 3: Compensate for reference voltage differences.

The off_ana term represents the offset at the input of the ADC, meaning that the reference voltage will have an impact on the magnitude of the offset at the output. Production calibration values are determined with a 1.25 V reference source. If a voltage significantly different than 1.25 V is used for V_{REF}, adjust the off_ana term by a factor of 1.25 / V_{REF}.

$$\text{off_ana} = \text{off_ana} * (1.25 / V_{\text{REF}})$$

Step 4: Calculate total offset by adding the analog offset to the systematic offset.

Systematic offset is a fixed number dependent on OSR, and calculated according to the following equation:

$$\text{off_sys} = 640 * (256 / \text{OSR})$$

Total uncorrected offset (off_tot) is calculated by:

$$\text{off_tot} = (\text{off_ana} * 4 + \text{off_sys})$$

Step 5: Apply gain error correction, if needed.

Before writing the OFFSET field, the total uncorrected offset must be multiplied by the gain calibration factor. If the gain calibration factor is equal to 1.0 (0x8000 in 16-bit hex, or GAIN3MSB = 1 and GAIN13LSB = 0), this step may be skipped. Otherwise, adjust off_tot according to the following equation:

$$\text{off_tot} = \text{GAIN_FACTOR} * (\text{off_tot} + 0x80000) - 0x80000$$

where $\text{GAIN_FACTOR} = \text{GAINCANAn} / 32768$.

Step 6: Write the offset correction value to the OFFSET field.

The OFFSET field holds an 18-bit 2's complement number, which should be the negation of the total offset, or -(off_tot). Before writing to the SCALE register, any leading sign bits should be masked off to avoid corrupting the programmed gain settings.

$$\text{OFFSET} = 0x3FFFF \& (-\text{off_tot})$$

23.3.6.1.2.2 Offset Correction in High Speed Mode

Offset correction for High Speed mode is identical to the procedure for Normal mode, with the exception of the calibration terms used in the DEVINFO space. The same OFFSETANABASE term is used, but OFFSETANA1HISPD, OFFSETANA2HISPD and OFFSETANA3HISPD are used instead of the OFFSETANAxNORM values. Refer to [23.3.6.1.2.1 Offset Correction in Normal Mode](#) for the procedure, replacing OFFSETANAxNORM with OFFSETANAxHISPD.



23.3.6.1.2.3 Offset Correction in High Accuracy Mode

The production offset calibration for High Accuracy mode uses two 16-bit terms written to the DEVINFO space: OFFSETANA1HIACC and OFFSETANABASE. The following procedure will determine the setting for the OFFSET register based on production calibration values.

Step 1: Calculate the analog offset adjustment term (off_ana) based on the OSR setting (OSRHA in IADC_CFGn).

off_ana = OFFSETANABASE + OFFSETANA1HIACC / (2^{OSRHA})

The following table expresses this relationship:

Table 23.8. Analog Offset Adjustment

OSRHA Setting	OSR	Analog Offset Adjustment Term (off_ana)
HIACC16	16 x	OFFSETANABASE + OFFSETANA1HIACC
HIACC32	32 x	OFFSETANABASE + (OFFSETANA1HIACC / 2)
HIACC64	64 x	OFFSETANABASE + (OFFSETANA1HIACC / 4)
HIACC92	92 x	OFFSETANABASE + (OFFSETANA1HIACC / 8)
HIACC128	128 x	OFFSETANABASE + (OFFSETANA1HIACC / 16)
HIACC256	256 x	OFFSETANABASE + (OFFSETANA1HIACC / 32)

Step 2: Compensate for reference voltage differences.

The off_ana term represents the offset at the input of the ADC, meaning that the reference voltage will have an impact on the magnitude of the offset at the output. Production calibration values are determined with a 1.25 V reference source. If a voltage significantly different than 1.25 V is used for V_{REF}, adjust the off_ana term by a factor of 1.25 / V_{REF}.

off_ana = off_ana * (1.25 / V_{REF})

Step 3: Calculate total offset by adding the analog offset to the systematic offset.

Systematic offset is a fixed number dependent on OSR, and calculated according to the following equation:

off_sys = 0x40000 / (OSR / (OSR+1))

Total uncorrected offset (off_tot) is calculated by:

off_tot = (off_ana * 4 + off_sys)

Step 4: Apply gain error correction.

Before writing the OFFSET field, the total uncorrected offset must be multiplied by the gain calibration factor according to the following equation:

off_tot = GAIN_FACTOR * (off_tot + 0x80000) - 0x80000

where GAIN_FACTOR = GAINCANAn / 32768.

Step 5: Write the offset correction value to the OFFSET field.

The OFFSET field holds an 18-bit 2's complement number, which should be the negation of the total offset, or -(off_tot). Before writing to the SCALE register, any leading sign bits should be masked off to avoid corrupting the programmed gain settings.

OFFSET = 0x3FFFF & -(off_tot)

23.3.6.2 Calibration

Calibration can be performed in-system to correct for external errors and provide more accurate measurements. The general calibration procedure is as follows:

1. Configure the ADC to the desired mode, OSR, analog gain settings, reference source, etc.
2. Force the IADC to use bipolar output for the conversion: `IADC_CFGx_TWOSCOMPL = FORCEBIPOLAR`.
3. Set the initial offset to the maximum negative value (`IADC_SCALEx_OFFSET = 0x20000`), and the initial gain to 1.0 (`GAIN3MSB = 1`, `GAIN13LSB = 0x0000`). This will prevent output saturation when measuring full scale.
4. Apply a full-scale positive input to the IADC and perform a conversion (*result_fullscale*). Multiple conversions can be performed and averaged together to reduce any system-level noise.
5. Apply a zero input to the IADC and perform a conversion (*result_zero*). Multiple conversions can be performed and averaged together to reduce any system-level noise.
6. Calculate the gain correction factor: Divide the expected value by the difference in the measured values (*result_fullscale* - *result_zero*). Note that the offset adjustment in Step 3 will be canceled out by this calculation.
7. Write the gain correction factor to the IADC using the `GAIN3MSB` and `GAIN13LSB` fields in `IADC_SCALEx`.
8. Set `IADC_SCALEx_OFFSET` to `0x00000` in preparation for the offset calibration.
9. Apply the desired zero voltage to the IADC input and perform a conversion (*result_offset*). Multiple conversions can be performed and averaged together to reduce any system-level noise.
10. Multiply *result_offset* to convert to a 20-bit value (*result_offset_20*). For example, a 12-bit result should be multiplied by 256.
11. Negate *result_offset_20* and write the value to `IADC_SCALEx_OFFSET`.

Note that the `IADC_SCALEx_OFFSET` field is 18 bits. If the result is greater than $(2^{17} - 1)$ or less than (-2^{17}) , the offset is too large to be corrected.

23.3.7 Output Data FIFOs

The single and scan queues each have a eight-word data FIFO. Conversions results are written to the output data FIFO associated with the queue. Single queue results are written to the single FIFO and scan queue results are written to the scan data FIFO. The two queues are identical in operation, but independent.

Conversion results are read from the single FIFO using `IADC_SINGLEFIFODATA`. Reading `SINGLEFIFODATA` will pop the oldest result from the FIFO. It is also possible to read the most recent valid data word using `IADC_SINGLEDATA`. Reading `SINGLEDATA` does not pop a conversion from the FIFO. Similarly, the scan FIFO results are read with `IADC_SCANFIFODATA`, which reads the oldest result and pops the FIFO. The most recent scan result can be read using `IADC_SCANDATA`.

When the single FIFO has valid data, the `SINGLEFIFODV` flag in `IADC_STATUS` is set to 1. When the scan FIFO has valid data `SCANFIFODV` in `IADC_STATUS` is set to 1. These data valid status bits are cleared automatically whenever the associated FIFO is empty. For more granular FIFO status, the number of data words present in the FIFO is indicated in `IADC_SINGLEFIFOSTAT` (for single FIFO) or `IADC_SCANFIFOSTAT` (for scan FIFO).

A programmable data level watermark is also available for the FIFOs, allowing hardware to trigger interrupts or DMA operations when a specified number of conversion results are available. The `DVL` field in register `SINGLEFIFOCFG` or `SCANFIFOCFG` sets the watermark level, between 1 and 4 conversions. If the number of valid entries in the FIFO reaches or exceeds the level set in `DVL`, the `SINGLEFIFODVLIF` (for single FIFO) or `SCANFIFODVLIF` (for scan FIFO) flag in the `IADC_IF` register will be set to 1. If enabled, an interrupt or DMA request will be triggered when the flag is set.

By default, DMA requests are turned off for operation in EM2 or EM3. However, the `DMAWUFIFOSINGLE` or `DMAWUFIFOSCAN` bits in `SINGLEFIFOCFG` or `SCANFIFOCFG` may be used to enable DMA operations in these lower energy modes.

Overflow and underflow status flags are also available in `IADC_IF`. An overflow condition occurs when an IADC conversion completes, but the associated FIFO is already full. In an overflow case the `SINGLEFIFOOFIF` or `SCANFIFOOFIF` flag will be set. The most recent conversion will still be available in the `SINGLEDATA` or `SCANDATA` register, but the FIFO will not be updated with the new data. An underflow condition occurs when software or hardware attempts to read from an empty FIFO. In an underflow case the `SINGLEFIFOUFIF` or `SCANFIFOUFIF` flag will be set.

23.3.7.1 Data Alignment and Channel ID

The IADC has data alignment options and the ability to include a channel ID along with the conversion data. For the single queue, alignment and channel ID are configured in the IADC_SINGLEFIFOCFG register. For the scan queue, alignment and channel ID are configured in the IADC_SCANFIFOCFG register.

The ALIGNMENT bit field specifies the data justification and the number of data bits as shown in [Figure 23.19 Data Alignment on page 771](#). By default, the converter will produce 12-bit right-justified data, corresponding to ALIGNMENT = RIGHT12.

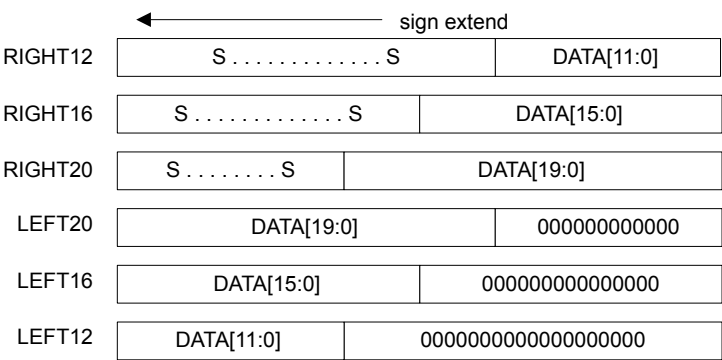


Figure 23.19. Data Alignment

The SHOWID bit controls whether the conversion channel ID is included in the output data word. This option is primarily used with the scan FIFO to help software determine which channel each conversion result came from. If SHOWID is enabled for single conversions, the ID will always be set to 0x20. [Figure 23.20 Data Alignment With ID on page 771](#) shows output data formatting including the ID, when SHOWID = 1.

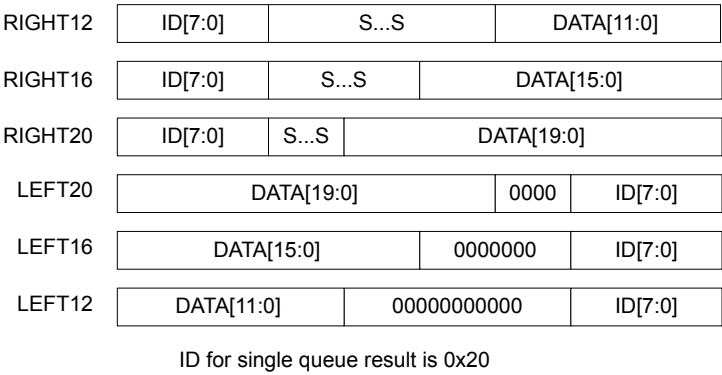


Figure 23.20. Data Alignment With ID

23.3.7.2 Output Polarity

The output polarity of the IADC is controlled by the TWOSCOMPL field in the IADC_CFGx register. The IADC supports unipolar and bipolar output formatting independent of the input configuration. By default, the TWOSCOMPL field is set to AUTO, meaning that single-ended conversions will produce unipolar output, and differential conversions will produce bipolar output. The polarity can be forced to unipolar or bipolar mode by setting TWOSCOMPL to FORCEUNIPOLAR or FORCEBIPOLAR, respectively.

Unipolar samples are unsigned integers representing zero to positive full-scale. Bipolar samples are two's-complement signed integers, representing negative full-scale to positive full-scale. Using unipolar mode on a differential input signal allows for more dynamic range when the signal is positive, but will saturate to zero when the signal is negative.

Note: If bipolar output is used with a single-ended input configuration, it is possible to see negative output values when the input is close to ground. However, the input voltage is still limited by the supply range of the device.

23.3.7.3 Digital Accumulation and Averaging

The IADC may optionally accumulate and average several conversion results before posting an output word to the FIFO. Digital averaging is controlled by the DIGAVG field in the IADC_CFGx register. It can be configured to average 1, 2, 4, 8, or 16 samples. The IADC will collect the number of samples specified by DIGAVG on the selected channel slot back-to-back, and produce only one averaged output word.

23.3.7.4 Output Resolution

The usable output resolution of the IADC is a minimum of 12 bits, when the oversampling ratio is set to 2 and no digital averaging is used (DIGAVG = AVG1). An extra bit of output resolution is produced for every power of 2 increase in either of these settings. In other words, the output resolution of the ADC can be determined as:

$$\text{Output Resolution} = 11 + \log_2(\text{OversamplingRatio} \times \text{DigitalAveraging})$$

The MSB is always left-aligned within the DATA field, and the output word will be truncated to 12, 16, or 20 bits, as shown in [Figure 23.19 Data Alignment on page 771](#) and [Figure 23.20 Data Alignment With ID on page 771](#). When using 16 or 20 bit alignment with lower oversampling ratio and digital averaging settings, LSBs of the output can contain residual effects of the offset and gain computation. These residual effects do not represent additional information about the input signal. Any extra LSBs can be masked to 0 by software.

Table 23.9. Output Resolution Masking Examples

Alignment Setting	Oversampling Ratio	Digital Averaging	Number of averaged samples	Output Resolution	Recommended Mask for DATA field
16-bit	2x	1x	2	12 bits	0xFFF0
16-bit	8x	2x	16	15 bits	0xFFFE
20-bit	2x	1x	2	12 bits	0xFFF00
20-bit	16x	4x	64	17 bits	0xFFFF8

23.3.7.5 Flushing the FIFOs

Each FIFO has a command bit in the IADC_CMD register that can be used to trigger a FIFO flush. The FIFO data may be flushed independently for each queue. To flush a FIFO:

1. The IADC must be enabled with the clock running.
2. Disable the queue associated with the FIFO using the SCANSTOP or SINGLESTOP bits in the IADC_CMD register.
3. Ensure the queue is disabled by reading the associated flag in the IADC_STATUS register (SINGLEQEN or SCANQEN).
4. Set the command bit to flush the desired FIFO (SINGLEFIFOFLUSH or SCANFIFOFLUSH) in the IADC_CMD register.
5. Wait for the corresponding status bit (SINGLEFIFOFLUSHING or SCANFIFOFLUSHING) in IADC_STATUS to go low.

23.3.8 Window Compare

The IADC has a window comparison unit that can trigger interrupts conditional on the output data of the converter. The window comparison unit has two thresholds - greater than or equal (ADGT), and less than or equal (ADLT), which are programmable through the IADC_CMPTHR register. The ADGT and ADLT thresholds always use a 16 bit, left-justified format, regardless of the format specified by the FIFO. The 12-bit conversion result will be compared against the upper 12 bits of the window comparator.

The window comparison unit is active on the ADC output on a conversion-by-conversion basis, and is shared between the two FIFOs. It is not possible to set different window comparison thresholds for different channels or for each FIFO. However, each channel specified in the IADC has a CMP bit field to enable the window comparison on results from that channel. For example, it is possible to only apply the window comparison and associated interrupt to scan channel #3 by setting the CMP field in IADC_SCAN3 to 1. When the CMP field associated with a channel is 0, the window comparator will not be active for results from that channel.

The window comparator supports conditional triggering on output results which are inside or outside a specified window. When ADLT is greater than or equal to ADGT, the comparator will trigger on an "inside" condition, or when $DATA \leq ADLT$ and $DATA \geq ADGT$. When ADLT is less than ADGT, the comparator will trigger on an "outside" condition, or when $DATA \leq ADLT$ or $DATA \geq ADGT$.

Figure 23.21 Window Comparison Examples on page 773 shows different configurations of the ADLT and ADGT values and the resulting windows. When the window comparator detects that the appropriate conditions are met (shown by the shaded region in the figure), it will generate an interrupt via the SINGLECMPIF flag for conversions on the single queue, or via the SCANCMPIF flag for conversions on the scan queue.

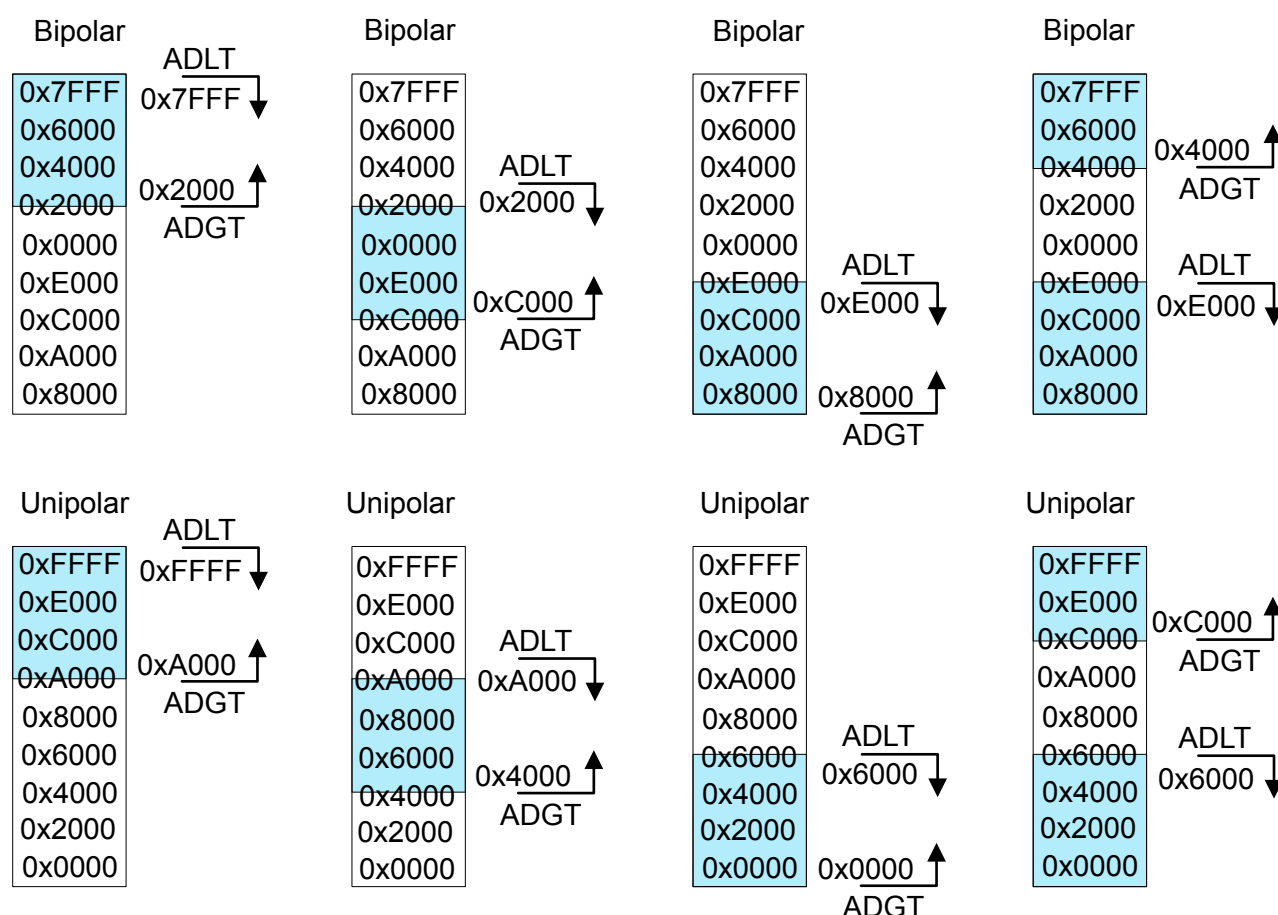


Figure 23.21. Window Comparison Examples

23.3.9 Interrupts

Interrupts are enabled in the IADC_IEN register, allowing interrupts to be generated on several different IADC conditions. Each of the flags in IADC_IF has a corresponding enable bit in the IADC_IEN register. A brief overview of the available interrupt sources is shown in the list below; more details can be found in the relevant sections of this chapter.

- SINGLEFIFODVLIF - The single FIFO watermark specified in SINGLEFIFOCFG_DVL has been reached or exceeded.
- SCANFIFODVLIF - The scan FIFO watermark specified in SCANFIFOCFG_DVL has been reached or exceeded.
- SINGLECMPIF - A conversion result from the single queue tripped the window comparator.
- SCANCMPIF - A conversion result from the scan queue tripped the window comparator.
- SCANENTRYDONEIF - A scan queue conversion has completed.
- SCANTABLEDONEIF - A scan queue operation has completed (all channels specified in the scan mask have been converted once).
- POLARITYERRIF - A channel polarity selection error has occurred (two channels from the EVEN multiplexer or two channels from the ODD multiplexer were selected for positive and negative inputs).
- PORTALLOCERRIF - A port allocation error has occurred (a pin not allocated to the IADC in the GPIO bus allocation registers was requested).
- SINGLEFIFOOFIF - A single FIFO overflow has occurred.
- SCANFIFOOFIF - A scan FIFO overflow has occurred.
- SINGLEFIFOUFIF - A single FIFO underflow has occurred.
- SCANFIFOUFIF - A scan FIFO underflow has occurred.
- EM23ABORTERRORIF - The system entered EM2 or EM3 while the IADC was converting and using a clock not supported in EM2 or EM3.

Hardware sets the interrupt flags in IADC_IF, and the flags remain set (sticky) until cleared by software. The interrupts flags should be cleared before enabling the IADC to remove any previous interrupt history. Clearing or setting interrupt bits can be done by writing to IADC_IF with a set or clear mask.

23.4 IADC Register Map

The offset register address is relative to the registers base address.

Offset	Name	Type	Description
0x000	IADC_IPVERSION	R	IPVERSION
0x004	IADC_EN	RW ENABLE	Enable
0x008	IADC_CTRL	RW CONFIG	Control
0x00C	IADC_CMD	W SYNC	Command
0x010	IADC_TIMER	RW CONFIG	Timer
0x014	IADC_STATUS	RH	Status
0x018	IADC_MASKREQ	RW SYNC	Mask Request
0x01C	IADC_STMASK	RH SYNC	Scan Table Mask
0x020	IADC_CMPTHR	RW CONFIG	Digital Window Comparator Threshold
0x024	IADC_IF	RWH INTFLAG	Interrupt Flags
0x028	IADC_IEN	RW	Interrupt Enable
0x02C	IADC_TRIGGER	RW CONFIG	Trigger
0x048	IADC_CFGx	RW CONFIG	Configuration
0x050	IADC_SCALEx	RW CONFIG	Scaling
0x054	IADC_SCHEx	RW CONFIG	Scheduling
0x070	IADC_SINGLEFIFOCFG	RW CONFIG	Single FIFO Configuration
0x074	IADC_SINGLEFIFODATA	RH(r)	Single FIFO DATA
0x078	IADC_SINGLEFIFOSTAT	RH	Single FIFO Status
0x07C	IADC_SINGLEDATA	RH SYNC	Single Data
0x080	IADC_SCANFIFOCFG	RW CONFIG	Scan FIFO Configuration
0x084	IADC_SCANFIFODATA	RH(r)	Scan FIFO Read Data
0x088	IADC_SCANFIFOSTAT	RH	Scan FIFO Status
0x08C	IADC_SCANDATA	RH SYNC	Scan Data
0x098	IADC_SINGLE	RW SYNC	Single Queue Port Selection
0x0A0	IADC_SCANx	RW CONFIG	SCAN Entry
0x1000	IADC_IPVERSION_SET	R	IPVERSION
0x1004	IADC_EN_SET	RW ENABLE	Enable
0x1008	IADC_CTRL_SET	RW CONFIG	Control
0x100C	IADC_CMD_SET	W SYNC	Command
0x1010	IADC_TIMER_SET	RW CONFIG	Timer
0x1014	IADC_STATUS_SET	RH	Status
0x1018	IADC_MASKREQ_SET	RW SYNC	Mask Request
0x101C	IADC_STMASK_SET	RH SYNC	Scan Table Mask
0x1020	IADC_CMPTHR_SET	RW CONFIG	Digital Window Comparator Threshold
0x1024	IADC_IF_SET	RWH INTFLAG	Interrupt Flags

Offset	Name	Type	Description
0x1028	IADC_IEN_SET	RW	Interrupt Enable
0x102C	IADC_TRIGGER_SET	RW CONFIG	Trigger
0x1048	IADC_CFGx_SET	RW CONFIG	Configuration
0x1050	IADC_SCALEx_SET	RW CONFIG	Scaling
0x1054	IADC_SCHEx_SET	RW CONFIG	Scheduling
0x1070	IADC_SINGLEFIFOCFG_SET	RW CONFIG	Single FIFO Configuration
0x1074	IADC_SINGLEFIFODATA_SET	RH(r)	Single FIFO DATA
0x1078	IADC_SINGLEFIFOSTAT_SET	RH	Single FIFO Status
0x107C	IADC_SINGLEDATA_SET	RH SYNC	Single Data
0x1080	IADC_SCANFIFOCFG_SET	RW CONFIG	Scan FIFO Configuration
0x1084	IADC_SCANFIFODATA_SET	RH(r)	Scan FIFO Read Data
0x1088	IADC_SCANFIFOSTAT_SET	RH	Scan FIFO Status
0x108C	IADC_SCANDATA_SET	RH SYNC	Scan Data
0x1098	IADC_SINGLE_SET	RW SYNC	Single Queue Port Selection
0x10A0	IADC_SCANx_SET	RW CONFIG	SCAN Entry
0x2000	IADC_IPVERSION_CLR	R	IPVERSION
0x2004	IADC_EN_CLR	RW ENABLE	Enable
0x2008	IADC_CTRL_CLR	RW CONFIG	Control
0x200C	IADC_CMD_CLR	W SYNC	Command
0x2010	IADC_TIMER_CLR	RW CONFIG	Timer
0x2014	IADC_STATUS_CLR	RH	Status
0x2018	IADC_MASKREQ_CLR	RW SYNC	Mask Request
0x201C	IADC_STMASK_CLR	RH SYNC	Scan Table Mask
0x2020	IADC_CMPTHR_CLR	RW CONFIG	Digital Window Comparator Threshold
0x2024	IADC_IF_CLR	RWH INTFLAG	Interrupt Flags
0x2028	IADC_IEN_CLR	RW	Interrupt Enable
0x202C	IADC_TRIGGER_CLR	RW CONFIG	Trigger
0x2048	IADC_CFGx_CLR	RW CONFIG	Configuration
0x2050	IADC_SCALEx_CLR	RW CONFIG	Scaling
0x2054	IADC_SCHEx_CLR	RW CONFIG	Scheduling
0x2070	IADC_SINGLEFIFOCFG_CLR	RW CONFIG	Single FIFO Configuration
0x2074	IADC_SINGLEFIFODATA_CLR	RH(r)	Single FIFO DATA
0x2078	IADC_SINGLEFIFOSTAT_CLR	RH	Single FIFO Status
0x207C	IADC_SINGLEDATA_CLR	RH SYNC	Single Data
0x2080	IADC_SCANFIFOCFG_CLR	RW CONFIG	Scan FIFO Configuration
0x2084	IADC_SCANFIFODATA_CLR	RH(r)	Scan FIFO Read Data
0x2088	IADC_SCANFIFOSTAT_CLR	RH	Scan FIFO Status

Offset	Name	Type	Description
0x208C	IADC_SCANDATA_CLR	RH SYNC	Scan Data
0x2098	IADC_SINGLE_CLR	RW SYNC	Single Queue Port Selection
0x20A0	IADC_SCANx_CLR	RW CONFIG	SCAN Entry
0x3000	IADC_IPVERSION_TGL	R	IPVERSION
0x3004	IADC_EN_TGL	RW ENABLE	Enable
0x3008	IADC_CTRL_TGL	RW CONFIG	Control
0x300C	IADC_CMD_TGL	W SYNC	Command
0x3010	IADC_TIMER_TGL	RW CONFIG	Timer
0x3014	IADC_STATUS_TGL	RH	Status
0x3018	IADC_MASKREQ_TGL	RW SYNC	Mask Request
0x301C	IADC_STMASK_TGL	RH SYNC	Scan Table Mask
0x3020	IADC_CMPTHR_TGL	RW CONFIG	Digital Window Comparator Threshold
0x3024	IADC_IF_TGL	RWH INTFLAG	Interrupt Flags
0x3028	IADC_IEN_TGL	RW	Interrupt Enable
0x302C	IADC_TRIGGER_TGL	RW CONFIG	Trigger
0x3048	IADC_CFGx_TGL	RW CONFIG	Configuration
0x3050	IADC_SCALEx_TGL	RW CONFIG	Scaling
0x3054	IADC_SCHEx_TGL	RW CONFIG	Scheduling
0x3070	IADC_SINGLEFIFOCFG_TGL	RW CONFIG	Single FIFO Configuration
0x3074	IADC_SINGLEFIFODATA_TGL	RH(r)	Single FIFO DATA
0x3078	IADC_SINGLEFIFOSTAT_TGL	RH	Single FIFO Status
0x307C	IADC_SINGLEDATA_TGL	RH SYNC	Single Data
0x3080	IADC_SCANFIFOCFG_TGL	RW CONFIG	Scan FIFO Configuration
0x3084	IADC_SCANFIFODATA_TGL	RH(r)	Scan FIFO Read Data
0x3088	IADC_SCANFIFOSTAT_TGL	RH	Scan FIFO Status
0x308C	IADC_SCANDATA_TGL	RH SYNC	Scan Data
0x3098	IADC_SINGLE_TGL	RW SYNC	Single Queue Port Selection
0x30A0	IADC_SCANx_TGL	RW CONFIG	SCAN Entry

23.5 IADC Register Description

23.5.1 IADC_IPVERSION - IPVERSION

Offset	Bit Position																																					
0x000	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0						
Reset	0x3																																					
Access	R																																					
Name	IPVERSION																																					

Bit	Name	Reset	Access	Description
31:0	IPVERSION	0x3	R	IP version ID
				The read only IPVERSION field gives the version for this module. There may be minor software changes required for modules with different values of IPVERSION.

23.5.2 IADC_EN - Enable

Offset	Bit Position																																					
0x004	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0						
Reset																															0x0	0x0						
Access																															R	RW						
Name																															DISABLING	EN						

Bit	Name	Reset	Access	Description
31:2	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
1	DISABLING	0x0	R	Disablement busy status
				When EN is cleared, DISABLING status is set immediately, and cleared when disablement finishes. Disablement resets peripheral cores and not APB registers except hardware updated registers such as INTFLAGS and FIFOs.
0	EN	0x0	RW	Enable IADC Module
				The EN bit enables the module. Software should write to CONFIG type registers before setting the EN bit. Software should write to SYNC type registers only after setting the EN bit.
Value		Mode		Description
0		DISABLE		Disable
1		ENABLE		Enable

23.5.3 IADC CTRL - Control

Offset	Bit Position																																			
0x008	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0				
Reset			0x0							0x0																	0x0			0x0	2	1	0			
Access			RW							RW																	RW		RW	0x0	RW	0x0	RW	0x0		
Name		HSCLKRATE								TIMEBASE																	WARMUPMODE		DBGHALT		ADCCCLKSUSPEND1		ADCCCLKSUSPEND0		EM23WU_CONVERT	

Bit	Name	Reset	Access	Description
31	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
30:28	HSCLKRATE	0x0	RW	High Speed Clock Rate Ratio to divide incoming CLK_CMU_ADC clock by. The resulting clock (CLK_SRC_ADC) must be 40 MHz or less.
	Value	Mode	Description	
	0	DIV1	Use CMU_CLK_ADC directly. The source clock must be 40 MHz or less.	
	1	DIV2	Divide CMU_CLK_ADC by 2 before using it. The resulting CLK_SRC_ADC must be 40 MHz or less.	
	2	DIV3	Divide CMU_CLK_ADC by 3 before using it. The resulting CLK_SRC_ADC must be 40 MHz or less.	
	3	DIV4	Divide CMU_CLK_ADC by 4 before using it. The resulting CLK_SRC_ADC must be 40 MHz or less.	
27:23	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
22:16	TIMEBASE	0x0	RW	Time Base ADC clock cycles (TIMEBASE + 1) needed to generate a 1 us interval for warm up and start up timing. Does not allow less than 2 cycles. A setting of 0x0 (1 cycle) is replaced with 0x1 (2 cycles).
15:6	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
5:4	WARMUPMODE	0x0	RW	Warmup Mode Select the warmup mode for the ADC.
	Value	Mode	Description	
	0	NORMAL	Shut down the IADC after conversions have completed.	
	1	KEEPINSTANDBY	Switch to standby mode after conversions have completed. The next warmup time will require 1us.	
	2	KEEPWARM	Keep IADC fully powered after conversions have completed.	
3	DBGHALT	0x0	RW	Debug Halt

Bit	Name	Reset	Access	Description
	ADC behavior when halted by debugger.			
	Value	Mode		Description
	0	NORMAL		Continue operation as normal during debug mode
	1	HALT		Complete the current conversion and then halt during debug mode
2	ADCCLKSUSPEND1	0x0	RW	ADC_CLK Suspend - PRS1 This only functions with single trigger select set to PRSPOS or PRSNEG. In EM0 and EM1, this gates the local clock while clock source remains running. In EM2 and EM3, this disables the clock source until the PRSPOS or PRSNEG event is detected. This bit has no effect if the local IADC timer is running.
	Value	Mode		Description
	0	PRSWUDIS		Normal mode which does not disable the ADC_CLK.
	1	PRSWUEN		ADCCLKWUEN will gate off ADC_CLK until the trigger is detected provided the internal timer is not selected as the trigger. Once the trigger is detected the ADC_CLK will be started, the band gap will be started, the ADC will be warmed up, and the SCAN Table and the Single entry will be converted. Once the conversions are done, the ADC_CLK will be gated off.
1	ADCCLKSUSPEND0	0x0	RW	ADC_CLK Suspend - PRS0 This only functions with scan trigger select set to PRSPOS or PRSNEG. In EM0 and EM1, this gates the local clock while clock source remains running. In EM2 and EM3, this disables the clock source until the PRSPOS or PRSNEG event is detected. This bit has no effect if the local IADC timer is running.
	Value	Mode		Description
	0	PRSWUDIS		Normal mode which does not disable the ADC_CLK.
	1	PRSWUEN		ADCCLKWUEN will gate off ADC_CLK until the trigger is detected provided the internal timer is not selected as the trigger. Once the trigger is detected the ADC_CLK will be started, the band gap will be started, the ADC will be warmed up, and the SCAN Table and the Single entry will be converted. Once the conversions are done, the ADC_CLK will be gated off.
0	EM23WUCONVERT	0x0	RW	EM23 Wakeup on Conversion EM23 wake up on conversion
	Value	Mode		Description
	0	WUDVL		When using suspend mode, conversions performed in EM2 or EM3 should not wake up the DMA until the FIFO's DVL setting is reached. This saves more power for large OSR settings or infrequent sampling.
	1	WUCONVERT		When using suspend mode, conversions performed in EM2 or EM3 will wake up the DMA and keep it awake until the conversions are done, regardless of the DVL setting. This mode burns more power, but it is useful when the conversion rate is faster than the time for the DMA to cycle through wake up and going back to sleep as it converts more than 4 scan table entries. Without using the wake up on conversion mode, the FIFO may overflow while the DMA is going in and out of sleep.

23.5.4 IADC_CMD - Command

Offset	Bit Position																																
0x00C	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
Reset							0x0	0x0									0x0	0x0											0x0	0x0		0x0	0x0
Access							W(nB)	W(nB)									W(nB)	W(nB)											W(nB)	W(nB)		W(nB)	W(nB)
Name							SCANFIFOFLUSH	SINGLEFIFOFLUSH									TIMERDIS	TIMEREN											SCANSTOP	SCANSTART		SINGLESTOP	SINGLESTART

Bit	Name	Reset	Access	Description
31:26	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
25	SCANFIFOFLUSH	0x0	W(nB)	Flush the Scan FIFO Flush the Scan FIFO. The IADC must be enabled, not suspended, and the IADC clock must be running. Operation has completed when STATUS.SCANFIFOFLUSHING has gone low. The scan queue should be disabled. Any incoming scan queue data will be discarded during the flush.
24	SINGLEFIFOFLUSH	0x0	W(nB)	Flush the Single FIFO Flush the Single FIFO. The IADC must be enabled, not suspended, and the IADC clock must be running. Operation has completed when STATUS.SINGLEFIFOFLUSHING has gone low. The Single queue should be disabled. Any incoming single queue data will be discarded during the flush.
23:18	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
17	TIMERDIS	0x0	W(nB)	Timer Disable Disable the local timer and reset the counter to timer reload value.
16	TIMEREN	0x0	W(nB)	Timer Enable Enable the local timer.
15:5	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
4	SCANSTOP	0x0	W(nB)	Scan Queue Stop Stop the Scan queue. Disables Scan triggers and clears pending conversions in the Scan queue. Any conversion that has already started will continue until it is complete. If the scan queue is stopped before all entries of the scan table have completed, the remaining entries will not be converted.
3	SCANSTART	0x0	W(nB)	Scan Queue Start Start the Scan queue. Enables triggering of the Scan queue.
2	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
1	SINGLESTOP	0x0	W(nB)	Single Queue Stop Stop the Single queue. Disables Single queue triggers and clears pending conversions in the Single queue. Any conversion that has already started will continue until it is complete.
0	SINGLESTART	0x0	W(nB)	Single Queue Start

Bit	Name	Reset	Access	Description
				Start the Single queue. Enables triggering of the Single queue.

23.5.5 IADC_TIMER - Timer

Offset	Bit Position																															
0x010	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reset																	0x0															
Access																	RW															
Name																	TIMER															

Bit	Name	Reset	Access	Description
31:16	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
15:0	TIMER	0x0	RW	Timer Period Number of CLK_SRC_ADC cycles per timer event.

23.5.6 IADC_STATUS - Status

Offset	Bit Position																																					
0x014	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0						
Reset		0x0						0x0				0x0	0x0				0x0	0x0	0x0	0x0				0x0	0x0		0x0	0x0	0x0	0x0	0x0	0x0						
Access		R						R				R	R				R	R	R				R	R		R	0x0		R	0x0	R	0x0	R	0x0	R			
Name		ADCWARM						SYNCBUSY				MASKREQWRITEPENDING	SINGLEWRITEPENDING				TIMERACTIVE	SCANFIFOFLUSHING	SINGLEFIFOFLUSHING				SCANFIFODV	SINGLEFIFODV				CONVERTING				SCANQUEUEPENDING	SCANQEN				SINGLEQUEUEPENDING	SINGLEQEN

Bit	Name	Reset	Access	Description
31	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
30	ADCWARM	0x0	R	ADCWARM The ADC analog front end and reference require a delay before converting when coming from a powered down or stand-by state. This status bit indicates that the analog front end and reference are ready.
29:25	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
24	SYNCBUSY	0x0	R	SYNCBUSY Indicates synchronization ongoing
23:22	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
21	MASKREQWRITE-PENDING	0x0	R	MASKREQ write pending A write to MASKREQ is pending. The ADC converts using a local working mask register, and only transfers MASKREQ to the local working version when the SCAN queue is not converting.
20	SINGLEWRITEPEND-ING	0x0	R	SINGLE write pending The SINGLE register write is pending. The ADC converts using a local working version of the SINGLE register, and only transfer SINGLE to the local working version when the SINGLE queue is not being converted.
19:17	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
16	TIMERACTIVE	0x0	R	Timer Active The local timer is running.
15	SCANFIFOFLUSHING	0x0	R	The Scan FIFO is flushing A scan data FIFO flush operation is in progress.
14	SINGLEFIFOFLUSHING	0x0	R	The Single FIFO is flushing A single data FIFO flush operation is in progress.

Bit	Name	Reset	Access	Description
13:10	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
9	SCANFIFODV	0x0	R	SCANFIFO Data Valid At least one result in the scan FIFO is ready to read.
8	SINGLEFIFODV	0x0	R	SINGLEFIFO Data Valid At least one result in the single FIFO is ready to read.
7	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
6	CONVERTING	0x0	R	Converting The ADC is warmed up and in the process of performing a conversion.
5	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
4	SCANQUEUEPENDING	0x0	R	Scan Queue Pending The Scan queue has been triggered and is waiting to start conversion.
3	SCANQEN	0x0	R	Scan Queued Enabled The Scan queue is enabled.
2	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
1	SINGLEQUEUEPENDING	0x0	R	Single Queue Pending The Single queue has been triggered and is waiting to start conversion. When tailgating is used, SINGLEQUEUEPENDING will remain high until the a scan operation has completed.
0	SINGLEQEN	0x0	R	Single Queue Enabled The Single queue is enabled.

23.5.7 IADC_MASKREQ - Mask Request

Offset	Bit Position																															
0x018	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reset																	0x0															
Access																	RW															
Name																	MASKREQ															

Bit	Name	Reset	Access	Description
31:16	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
15:0	MASKREQ	0x0	RW	Scan Queue Mask Request Allows software to specify which entries in the Scan table should be converted. For example MASKREQ = 0x8014 means that scan table entries 15, 4, and 2 will be converted. The other entries will not be converted.

23.5.8 IADC_STMASK - Scan Table Mask

Offset	Bit Position																															
0x01C	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reset																	0x0															
Access																	R															
Name																	STMASK															

Bit	Name	Reset	Access	Description
31:16	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
15:0	STMASK	0x0	R	Scan Table Mask This is the active / working copy of the MASKREQ register that the ADC uses. It will only be updated at the end of a scan sequence or when no scan is in progress.

23.5.9 IADC_CMPTHR - Digital Window Comparator Threshold

Offset	Bit Position																															
0x020	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reset	0x0																0x0															
Access	RW																RW															
Name	ADGT																ADLT															

Bit	Name	Reset	Access	Description
31:16	ADGT	0x0	RW	ADC Greater Than or Equal to Threshold Compare threshold value for greater-than or equal to comparison. ADGT should be specified in a left-justified, 16-bit format regardless of the FIFO ALIGNMENT setting. Comparisons with 12-bit formats will ignore the 4 LSBs of the ADGT value. Comparisons with 20-bit formats will ignore the 4 LSBs of the 20-bit result. Unipolar or bipolar mode is considered in the comparison. When ADGT is greater than ADLT, the comparison is true if the result is either greater than ADGT or less than ADLT, but false if the result falls between the values.
15:0	ADLT	0x0	RW	ADC Less Than or Equal to Threshold Compare threshold value for less-than or equal to comparison. ADLT should be specified in a left-justified, 16-bit format regardless of the FIFO ALIGNMENT setting. Comparisons with 12-bit formats will ignore the 4 LSBs of the ADLT value. Comparisons with 20-bit formats will ignore the 4 LSBs of the 20-bit result. Unipolar or bipolar mode is considered in the comparison. When ADGT is greater than ADLT, the comparison is true if the result is either greater than ADGT or less than ADLT, but false if the result falls between the values.

23.5.10 IADC_IF - Interrupt Flags

Offset	Bit Position																																				
0x024	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0					
Reset	0x0													0x0	0x0	0x0	0x0			0x0	0x0			0x0	0x0	0x0				0x0	0x0	0x0	0x0				
Access	RW													RW	RW	RW	RW			RW	RW			RW	RW	RW				RW	RW	RW	RW	RW	RW	RW	RW
Name	EM23ABORTERROR													SCANFIFOUF	SINGLEFIFOUF	SCANFIFOOF	SINGLEFIFOOF			PORTALLOCERR	POLARITYERR			SINGLEDONE	SCANTABLEDONE	SCANENTRYDONE				SCANCMP	SINGLECMP	SCANFIFODVL	SINGLEFIFODVL				

Bit	Name	Reset	Access	Description
31	EM23ABORTERROR	0x0	RW	EM2/3 Abort Error The system entered EM2 or EM3 during a conversion with an unsupported clock. Conversion results may be corrupted.
30:20	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
19	SCANFIFOUF	0x0	RW	Scan FIFO Underflow A scan FIFO underflow has occurred.
18	SINGLEFIFOUF	0x0	RW	Single FIFO Underflow A single FIFO underflow has occurred.
17	SCANFIFOOF	0x0	RW	Scan FIFO Overflow A scan FIFO overflow has occurred.
16	SINGLEFIFOOF	0x0	RW	Single FIFO Overflow A single FIFO overflow has occurred.
15:14	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
13	PORTALLOCERR	0x0	RW	Port Allocation Error A pin was selected on a port which has not been allocated to the IADC in GPIO control.
12	POLARITYERR	0x0	RW	Polarity Error Either two even channels or two odd channels were programmed into the channel mux selection. The ADC result will be set to 0xFFFF.
11:10	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
9	SINGLEDONE	0x0	RW	Single Conversion Done A single conversion has completed.
8	SCANTABLEDONE	0x0	RW	Scan Table Done A scan sequence completed. Set at the end of a scan sequence after all valid entries of the scan table have completed.
7	SCANENTRYDONE	0x0	RW	Scan Entry Done A scan table conversion completed. Set at the completion of each valid entry of the scan table.

Bit	Name	Reset	Access	Description
6:4	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
3	SCANCMP	0x0	RW	Scan Result Window Compare Scan digital compare window tripped.
2	SINGLECMP	0x0	RW	Single Result Window Compare Single digital compare window tripped.
1	SCANFIFODVL	0x0	RW	Scan FIFO Data Valid Level A minimum of (DVL+1) entries are ready to be read from the Scan FIFO.
0	SINGLEFIFODVL	0x0	RW	Single FIFO Data Valid Level A minimum of (DVL+1) entries are ready to be read from the Single FIFO.

23.5.11 IADC_IEN - Interrupt Enable

Offset	Bit Position																																			
0x028	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0				
Reset	0x0													0x0	0x0	0x0	0x0			0x0	0x0			0x0	0x0	0x0				0x0	0x0	0x0	0x0			
Access	RW													RW	RW	RW	RW			RW	RW			RW	RW	RW				RW	RW	RW	RW	RW	RW	RW
Name	EM23ABORTERR													SCANFIFO	SINGLEFIFO	SCANFIFO	SINGLEFIFO			PORTALCERR	POLARITYERR			SINGLEDONE	SCANTABLEDONE	SCANENTRYDONE				SCANCMP	SINGLECMP	SCANFIFOVL	SINGLEFIFOVL			

Bit	Name	Reset	Access	Description
31	EM23ABORTERROR	0x0	RW	EM2/3 Abort Error Enable EM2/3 Abort Error Enable
30:20	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
19	SCANFIFOUF	0x0	RW	Scan FIFO Underflow Enable Scan FIFO underflow interrupt enable
18	SINGLEFIFOUF	0x0	RW	Single FIFO Underflow Enable Single FIFO underflow interrupt enable
17	SCANFIFOOF	0x0	RW	Scan FIFO Overflow Enable Scan FIFO overflow interrupt enable
16	SINGLEFIFOOF	0x0	RW	Single FIFO Overflow Enable Single FIFO overflow interrupt enable
15:14	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
13	PORTALLOCERR	0x0	RW	Port Allocation Error Enable Port Allocation Error Enable
12	POLARITYERR	0x0	RW	Polarity Error Enable Polarity Error Enable
11:10	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
9	SINGLEDONE	0x0	RW	Single Conversion Done Enable Single Conversion Done interrupt enable
8	SCANTABLEDONE	0x0	RW	Scan Table Done Enable Scan Table Done interrupt enable
7	SCANENTRYDONE	0x0	RW	Scan Entry Done Enable Scan Entry Done interrupt enable

Bit	Name	Reset	Access	Description
6:4	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
3	SCANCMP Scan Result Window Compare Enable	0x0	RW	Scan Result Window Compare Enable
2	SINGLECMP Single Result Window Compare Enable	0x0	RW	Single Result Window Compare Enable
1	SCANFIFODVL Scan FIFO Data Valid Level interrupt enable	0x0	RW	Scan FIFO Data Valid Level Enable
0	SINGLEFIFODVL Single FIFO Data Valid Level interrupt enable	0x0	RW	Single FIFO Data Valid Level Enable

23.5.12 IADC_TRIGGER - Trigger

Offset	Bit Position																			
0x02C	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12
Reset																0x0				0x0
Access																RW				RW
Name																SINGLETAILGATE				SINGLETRIGACTION
																				</

Bit	Name	Reset	Access	Description
	0	IMMEDIATE		Immediate triggering. The single queue will be disabled once the conversion is complete, unless TRIGGERACTION is set to continuous.
	1	TIMER		Triggers when the local timer count reaches zero.
	2	PRSCLKGRP		Triggers on PRS1 from a timer module that is using the same clock group as the ADC and has been programmed to use the same clock source as the ADC. The prescale may be different between the ADC and the timer module.
	3	PRSPOS		Triggers on asynchronous PRS1 positive edge. Requires PRS1 to go low for 3 ADC_CLKs before another positive edge can be detected. Generates an additional delay of 1 to 2 CLK_SRC_ADC cycles for synchronization.
	4	PRSNEG		Triggers on asynchronous PRS1 negative edge. Requires PRS1 to go high for 3 ADC_CLKs before another negative edge can be detected. Generates an additional delay of 1 to 2 CLK_SRC_ADC cycles for synchronization. PRSNEG should only be used when the trigger source is from a module that remains powered during EM23. For modules (ie: TIMER) that power down during EM23, PRSPOS should be used for an asynchronous trigger, and PRSCLKGRP should be used for a synchronous trigger.
7:5	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
4	SCANTRIGACTION	0x0	RW	Scan Trigger Action Selects the trigger action for the scan queue.
	Value	Mode		Description
	0	ONCE		For TRIGSEL=IMMEDIATE, goes through the scan table once and disables queue. For TRIGSEL = TIMER, PRSCLKGRP, PRSPOS, PRSNEG, goes through the scan table once per trigger.
	1	CONTINUOUS		Goes through the scan table, converts each entry with a mask bit set, and puts it back into the scan queue to repeat again continuously. The queues are first come first serve. If both queues are triggered, the single queue will get to convert after each scan table completes. The scan queue will get to convert after each single conversion completes.
3	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
2:0	SCANTRIGSEL	0x0	RW	Scan Trigger Select Selects the trigger source for the scan queue.
	Value	Mode		Description
	0	IMMEDIATE		Immediate triggering. The scan queue will be disabled once all conversions in the scan table are complete, unless TRIGGERACTION is set to continuous.
	1	TIMER		Triggers when the local timer count reaches zero.
	2	PRSCLKGRP		Triggers on PRS0 from a timer module that is using the same clock group as the ADC and has been programmed to use the same clock source as the ADC. The prescale may be different between the ADC and the timer module.

Bit	Name	Reset	Access	Description
3		PRSPOS		Triggers on asynchronous PRS0 positive edge. Requires PRS0 to go low for 3 ADC_CLKs before another positive edge can be detected. Generates an additional delay of 1 to 2 CLK_SRC_ADC cycles for synchronization.
4		PRSNEG		Triggers on asynchronous PRS0 negative edge. Requires PRS0 to go high for 3 ADC_CLKs before another negative edge can be detected. Generates an additional delay of 1 to 2 CLK_SRC_ADC cycles for synchronization. PRSNEG should only be used when the trigger source is from a module that remains powered during EM23. For modules (ie: TIMER) that power down during EM23, PRSPOS should be used for an asynchronous trigger, and PRSCLKGRP should be used for a synchronous trigger.

23.5.13 IADC_CFGx - Configuration

Offset	Bit Position																																	
0x048	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
Reset			0x0						0x0				0x0				0x2								0x3				0x0		0x0			
Access			RW						RW						RW				RW								RW				RW			
Name			TWOSCOMPL						DIGAVG						REFSEL				ANALOGGAIN								OSRHA				OSRHS		ADCMODE	

Bit	Name	Reset	Access	Description
31:30	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
29:28	TWOSCOMPL	0x0	RW	Two's Complement
	Selects output word polarity.			
	Value	Mode		Description
	0	AUTO		Automatic: Single ended measurements are reported as unipolar and differential measurements are reported as bipolar.
	1	FORCEUNIPOLAR		Force all measurements to result in unipolar output. Negative differential numbers will saturate to 0.
2	FORCEBIPOLAR		Force all measurements to result in bipolar output. Single ended measurements are half the range, but allow for small negative measurements.	
27:24	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
23:21	DIGAVG	0x0	RW	Digital Averaging
	Number of output words to convert and average.			
	Value	Mode		Description
	0	AVG1		Collect one output word (no digital averaging).
	1	AVG2		Collect and average 2 digital output words.
	2	AVG4		Collect and average 4 digital output words.
	3	AVG8		Collect and average 8 digital output words.
4	AVG16		Collect and average 16 digital output words.	
20:19	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
18:16	REFSEL	0x0	RW	Reference Select
	Selects voltage reference.			
	Value	Mode		Description
0	VBGR		Internal 1.21 V reference.	

Bit	Name	Reset	Access	Description
	1	VREF		External Reference. (Calibrated for 1.25V nominal.)
	2	VREF2P5		External Reference. Supports 2.5V in high accuracy mode.
	3	VDDX		AVDD (unbuffered)
	4	VDDX0P8BUF		AVDD (buffered) * 0.8
15	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
14:12	ANALOGGAIN	0x2	RW	Analog Gain Sets analog front end gain.
	Value	Mode		Description
	1	ANAGAIN0P5		Analog gain of 0.5x.
	2	ANAGAIN1		Analog gain of 1x.
	3	ANAGAIN2		Analog gain of 2x.
	4	ANAGAIN3		Analog gain of 3x.
	5	ANAGAIN4		Analog gain of 4x.
11:8	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
7:5	OSRHA	0x3	RW	High Accuracy OSR Over sampling ratio for high accuracy conversions.
	Value	Mode		Description
	0	HIACC16		High accuracy over sampling of 16x.
	1	HIACC32		High accuracy over sampling of 32x.
	2	HIACC64		High accuracy over sampling of 64x.
	3	HIACC92		High accuracy over sampling of 92x.
	4	HIACC128		High accuracy over sampling of 128x.
	5	HIACC256		High accuracy over sampling of 256x.
4:2	OSRHS	0x0	RW	High Speed OSR Over sampling ratio for high speed conversions.
	Value	Mode		Description
	0	HISPD2		High speed over sampling of 2x.
	1	HISPD4		High speed over sampling of 4x.
	2	HISPD8		High speed over sampling of 8x.
	3	HISPD16		High speed over sampling of 16x.
	4	HISPD32		High speed over sampling of 32x.
	5	HISPD64		High speed over sampling of 64x.
1:0	ADCMODE	0x0	RW	ADC Mode Selects ADC conversion mode.

Bit	Name	Reset	Access	Description
	Value	Mode		Description
0		NORMAL		High speed mode with a maximum ADC_CLK of 10 MHz.
1		HIGHSPEED		Double high speed mode with a maximum ADC_CLK of 20 MHz. Power consumption is boosted to allow faster conversions.
2		HIGHACCURACY		High accuracy mode with maximum ADC_CLK of 5 MHz.

23.5.14 IADC_SCALEx - Scaling

Offset	Bit Position																															
0x050	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reset	0x1	0x0														0x2C000																
Access	RW	RW														RW																
Name	GAIN3MSB	GAIN13LSB														OFFSET																

Bit	Name	Reset	Access	Description
31	GAIN3MSB	0x1	RW	Gain 3 MSBs 3 MSBs of the 16-bit gain value (0=011 or 0.75; 1=1xx or 1.00). Example {GAIN3MSB, GAIN13LSB} = {100, 0_1001_0000_0000} = 1.07031x. Example {GAIN3MSB, GAIN13LSB} = {011, 0_0000_1010_0010} = 0.75494x.
	Value	Mode		Description
	0	GAIN011		Upper 3 bits of gain = 011 (0.75x)
	1	GAIN100		Upper 3 bits of gain = 100 (1.00x)
30:18	GAIN13LSB	0x0	RW	Gain 13 LSBs 13 LSBs of the 16-bit gain value.
17:0	OFFSET	0x2C000	RW	Offset Offset

23.5.15 IADC_SCHEDx - Scheduling

Offset	Bit Position																															
0x054	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reset																							0x0									
Access																							RW									
Name																							PRESCALE									

Bit	Name	Reset	Access	Description
31:10	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
9:0	PRESCALE	0x0	RW	Prescale Second level prescaler - divides the CLK_SRC_ADC by (PRESCALE + 1) to generate ADC_CLK. PRESCALE=0 should only be used with HSCLKRATE=0. (See text.)

23.5.16 IADC_SINGLEFIFOCFG - Single FIFO Configuration

Offset	Bit Position																			
0x070	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12
	11	10	9	8	7	6	5	4	3	2	1	0								
Reset													0x0				0x3		0x0	0x0
Access													RW				RW		RW	RW
Name													DMAWUFIFOSINGLE				DVL		SHOWID	ALIGNMENT

Bit	Name	Reset	Access	Description
31:9	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
8	DMAWUFIFOSINGLE	0x0	RW	Single FIFO DMA wakeup. Enables single FIFO to wake DMA in EM2 or EM3.
	Value	Mode	Description	
	0	DISABLED	While in EM2 or EM3, the DMA controller will not be requested.	
	1	ENABLED	While in EM2 or EM3, the DMA controller will be requested when the single FIFO reaches its Data Valid Level. [DVL must be set to 0 (VALID1).]	
7	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
6:4	DVL	0x3	RW	Data Valid Level Data valid level before requesting DMA transfer. If the number of words in the FIFO reaches or exceeds DVL+1, DMA requests will be generated.
	Value	Mode	Description	
	0	VALID1	When 1 entry in the single FIFO is valid, set the SINGLEFI-FODVL interrupt and request DMA.	
	1	VALID2	When 2 entries in the single FIFO are valid, set the SINGLEFI-FODVL interrupt and request DMA.	
	2	VALID3	When 3 entries in the single FIFO are valid, set the SINGLEFI-FODVL interrupt and request DMA.	
	3	VALID4	When 4 entries in the single FIFO are valid, set the SINGLEFI-FODVL interrupt and request DMA.	
	4	VALID5	When 5 entries in the single FIFO are valid, set the SINGLEFI-FODVL interrupt and request DMA.	
	5	VALID6	When 6 entries in the single FIFO are valid, set the SINGLEFI-FODVL interrupt and request DMA.	
	6	VALID7	When 7 entries in the single FIFO are valid, set the SINGLEFI-FODVL interrupt and request DMA.	

Bit	Name	Reset	Access	Description
	7	VALID8		When 8 entries in the single FIFO are valid, set the SINGLEFI-FODVL interrupt and request DMA.
3	SHOWID	0x0	RW	Show ID ID of 0x20 will be applied in the output words.
2:0	ALIGNMENT	0x0	RW	Alignment Alignment of output data written into FIFO.
	Value	Mode		Description
	0	RIGHT12		ID[7:0], SIGN_EXT, DATA[11:0]
	1	RIGHT16		ID[7:0], SIGN_EXT, DATA[15:0]
	2	RIGHT20		ID[7:0], SIGN_EXT, DATA[19:0]
	3	LEFT12		DATA[11:0], 000000000000, ID[7:0]
	4	LEFT16		DATA[15:0], 00000000, ID[7:0]
	5	LEFT20		DATA[19:0], 0000, ID[7:0]

23.5.17 IADC_SINGLEFIFODATA - Single FIFO DATA

Offset	Bit Position																															
0x074	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reset	0x0																															
Access	R(r)																															
Name	DATA																															

Bit	Name	Reset	Access	Description
31:0	DATA	0x0	R(r)	Single FIFO Read Data Reads and pops the oldest value from the single FIFO.

23.5.18 IADC_SINGLEFIFOSTAT - Single FIFO Status

Offset	Bit Position																															
0x078	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reset																											0x0					
Access																											R					
Name																											FIFOREADCNT					

Bit	Name	Reset	Access	Description
31:4	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
3:0	FIFOREADCNT	0x0	R	FIFO Read Count Number of valid entries available to read.

23.5.19 IADC_SINGLEDATA - Single Data

Offset	Bit Position																															
0x07C	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reset	0x0																															
Access	R																															
Name	DATA																															

Bit	Name	Reset	Access	Description
31:0	DATA	0x0	R	Data Reads the most recent data word from the single FIFO, but does not pop a value. Even if the FIFO has overflowed and stopped updating, the most recent conversion will continue to overwrite SINGLEDATA.

23.5.20 IADC_SCANFIFOCFG - Scan FIFO Configuration

Offset	Bit Position																															
0x080	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reset																								0x0		0x3			0x0		0x0	
Access																								RW		RW			RW		RW	
Name																								DMAWUFIFOSCAN		DVL			SHOWID		ALIGNMENT	

Bit	Name	Reset	Access	Description
31:9	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
8	DMAWUFIFOSCAN	0x0	RW	Scan FIFO DMA Wakeup Enables scan FIFO to wake DMA in EM2 or EM3.
	Value	Mode		Description
	0	DISABLED		While in EM2 or EM3, the DMA controller will not be requested.
	1	ENABLED		While in EM2 or EM3, the DMA controller will be requested when the scan FIFO reaches its Data Valid Level. [DVL must be set to 0 (VALID1).]
7	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
6:4	DVL	0x3	RW	Data Valid Level Data valid level before requesting DMA transfer. If the number of words in the FIFO reaches or exceeds DVL+1, DMA requests will be generated.
	Value	Mode		Description
	0	VALID1		When 1 entry in the scan FIFO is valid, set the SCANFIFODVL interrupt and request DMA.
	1	VALID2		When 2 entries in the scan FIFO are valid, set the SCANFIFODVL interrupt and request DMA.
	2	VALID3		When 3 entries in the scan FIFO are valid, set the SCANFIFODVL interrupt and request DMA.
	3	VALID4		When 4 entries in the scan FIFO are valid, set the SCANFIFODVL interrupt and request DMA.
	4	VALID5		When 5 entries in the scan FIFO are valid, set the SCANFIFODVL interrupt and request DMA.
	5	VALID6		When 6 entries in the scan FIFO are valid, set the SCANFIFODVL interrupt and request DMA.
	6	VALID7		When 7 entries in the scan FIFO are valid, set the SCANFIFODVL interrupt and request DMA.

Bit	Name	Reset	Access	Description
	7	VALID8		When 8 entries in the scan FIFO are valid, set the SCANFI-FODVL interrupt and request DMA.
3	SHOWID	0x0	RW	Show ID Enable ID in output words.
2:0	ALIGNMENT	0x0	RW	Alignment Alignment of output data written into FIFO.
	Value	Mode		Description
	0	RIGHT12		ID[7:0], SIGN_EXT, DATA[11:0]
	1	RIGHT16		ID[7:0], SIGN_EXT, DATA[15:0]
	2	RIGHT20		ID[7:0], SIGN_EXT, DATA[19:0]
	3	LEFT12		DATA[11:0], 000000000000, ID[7:0]
	4	LEFT16		DATA[15:0], 00000000, ID[7:0]
	5	LEFT20		DATA[19:0], 0000, ID[7:0]

23.5.21 IADC_SCANFIFODATA - Scan FIFO Read Data

Offset	Bit Position																															
0x084	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reset	0x0																															
Access	R(r)																															
Name	DATA																															

Bit	Name	Reset	Access	Description
31:0	DATA	0x0	R(r)	Data Reads and pops the oldest value from the scan FIFO.

23.5.22 IADC_SCANFIFOSTAT - Scan FIFO Status

Offset	Bit Position																															
0x088	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reset																											0x0					
Access																											R					
Name																											FIFOREADCNT					

Bit	Name	Reset	Access	Description
31:4	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
3:0	FIFOREADCNT	0x0	R	FIFO Read Count Number of valid entries available to read.

23.5.23 IADC_SCANDATA - Scan Data

Offset	Bit Position																															
0x08C	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reset	0x0																															
Access	R																															
Name	DATA																															

Bit	Name	Reset	Access	Description
31:0	DATA	0x0	R	Data Reads the most recent data word from the scan FIFO, but does not pop a value. Even if the FIFO has overflowed and stopped updating, the most recent conversion will continue to overwrite SCANDATA.

23.5.24 IADC_SINGLE - Single Queue Port Selection

Offset	Bit Position																															
0x098	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reset															0x0	0x0	0x0				0x0				0x0				0x0			
Access															RW	RW	RW				RW				RW				RW			
Name															CMP	CFG	PORTPOS				PINPOS				PORTNEG				PINNEG			

Bit	Name	Reset	Access	Description
31:18	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
17	CMP	0x0	RW	Comparison Enable Enable digital window comparison for this entry.
16	CFG	0x0	RW	Configuration Group Select Select which configuration group (CFGx, SCALEx, SCHEDx registers) is used with this entry.
	Value	Mode		Description
	0	CONFIG0		Use configuration group 0
	1	CONFIG1		Use configuration group 1
15:12	PORTPOS	0x0	RW	Positive Port Select Port (A, B, C, D) or special signal assigned to the positive input of the ADC
	Value	Mode		Description
	0	GND		Ground
	1	SUPPLY		Supply Pin - Select specific supply using PINPOS
	2	DAC0		Direct connection to DAC0_CH0
	4	PADANA0		Direct connection to AIN0 input pin
	5	PADANA2		Direct connection to AIN2 input pin
	8	PORTA		Port A - Select pin number using PINPOS
	9	PORTB		Port B - Select pin number using PINPOS
	10	PORTC		Port C - Select pin number using PINPOS
	11	PORTD		Port D - Select pin number using PINPOS
11:8	PINPOS	0x0	RW	Positive Pin Select Pin number for the positive input of the ADC.
7:4	PORTNEG	0x0	RW	Negative Port Select Port (A, B, C, D) or special signal assigned to the negative input of the ADC
	Value	Mode		Description
	0	GND		Ground (single-ended)

Bit	Name	Reset	Access	Description
	2	DAC1		Direct connection to DAC0_CH1
	4	PADANA1		Direct connection to AIN1 input pin
	5	PADANA3		Direct connection to AIN3 input pin
	8	PORTA		Port A - Select pin number using PINNEG
	9	PORTB		Port B - Select pin number using PINNEG
	10	PORTC		Port C - Select pin number using PINNEG
	11	PORTD		Port D - Select pin number using PINNEG
3:0	PINNEG	0x0	RW	Negative Pin Select Pin number for the negative input of the ADC.

23.5.25 IADC_SCANx - SCAN Entry

Offset	Bit Position															
0x0A0	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reset															0x0	0x0
Access															RW	RW
Name															CMP	CFG

Bit	Name	Reset	Access	Description
31:18	Reserved	To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in 1.2 Conventions		
17	CMP	0x0	RW	Comparison Enable Enable digital window comparison for this entry.
16	CFG	0x0	RW	Configuration Group Select Select which configuration group (CFGx, SCALEx, SCHEDx registers) is used with this entry.
	Value	Mode	Description	
	0	CONFIG0	Use configuration group 0	
	1	CONFIG1	Use configuration group 1	
15:12	PORTPOS	0x0	RW	Positive Port Select Port (A, B, C, D) or special signal assigned to the positive input of the ADC
	Value	Mode	Description	
	0	GND	Ground	
	1	SUPPLY	Supply Pin - Select specific supply using PINPOS	
	2	DAC0	Direct connection to DAC0_CH0	
	4	PADANA0	Direct connection to AIN0 input pin	
	5	PADANA2	Direct connection to AIN2 input pin	
	8	PORTA	Port A - Select pin number using PINPOS	
	9	PORTB	Port B - Select pin number using PINPOS	
	10	PORTC	Port C - Select pin number using PINPOS	
	11	PORTD	Port D - Select pin number using PINPOS	
11:8	PINPOS	0x0	RW	Positive Pin Select Pin number for the positive input of the ADC.
7:4	PORTNEG	0x0	RW	Negative Port Select Port (A, B, C, D) or special signal assigned to the negative input of the ADC
	Value	Mode	Description	
	0	GND	Ground (single-ended)	

Bit	Name	Reset	Access	Description
	2	DAC1		Direct connection to DAC0_CH1
	4	PADANA1		Direct connection to AIN1 input pin
	5	PADANA3		Direct connection to AIN3 input pin
	8	PORTA		Port A - Select pin number using PINNEG
	9	PORTB		Port B - Select pin number using PINNEG
	10	PORTC		Port C - Select pin number using PINNEG
	11	PORTD		Port D - Select pin number using PINNEG
3:0	PINNEG	0x0	RW	Negative Pin Select Pin number for the negative input of the ADC.